Industrial Component Anomaly Detection

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Abstract

This report documents our complete process of developing unsupervised anomaly detection models for industrial components, evaluated on the MVTec Anomaly Detection dataset. We describe the full trajectory of our work, from exploratory data analysis and iterative prototype experiments to a production-grade software architecture, and we explain each decision in the context of the findings that motivated it. Our final system comprises two complementary approaches: an advanced Keras convolutional autoencoder incorporating masked image modelling and a combined structural and pixel-wise loss function, and a PatchCore model with a ResNet-18 backbone. The choice of ResNet-18 was necessitated by the computational and memory constraints of consumer-grade hardware. Both systems are evaluated across all 15 categories using the same deterministic 85% training/15% validation split, with the official test set reserved for final evaluation. The primary metrics are the F1 score at image level and the Area Under the Per-Image Overlap (AUPIMO) at pixel level, complemented by precision-recall curves and threshold sensitivity curves.

1 Introduction and Task Description

This project was conducted as part of the Liora data science curriculum and with a difficulty rating of 10 out of 10. The objective was to develop a model capable of detecting anomalies in industrial components from images, using the MVTec AD dataset [1, 2]. This second report provides the technical account of the project, from data exploration to the evaluation of the completed system, and supplies the scientific basis for the subsequent business report. The task description explicitly references the paper *AUPIMO: Redefining Visual Anomaly Detection Benchmarks with High Speed and Low Tolerance* [5] as a key bibliographic source. This motivated our adoption of AUPIMO as the primary pixel-level evaluation metric, a choice we justify technically in Section 3.

Quality control in modern manufacturing operates under a fundamental constraint: defects are rare by design. A well-engineered production line yields thousands of good items for every defect. This scarcity makes supervised classification impractical in the classical sense, as there are simply not enough labelled defect examples to train a reliable closed-world classifier, and future defects take forms that were never seen during development. The standard response to this challenge is unsupervised anomaly detection. Our models train on defect-free normal images and learn compact representations of what normal looks like. At test time, any image that deviates significantly from this learned normal appearance is flagged as anomalous. This is both technically demanding and fundamentally different from standard classification, because the model must determine what is normal and anomalous from normal-only training data, with no exposure to any defect during training.

Throughout this report, we describe our work in the order in which we executed it, including our iterative challenges and course corrections. We believe this honest account of the learning process is as informative as the final results.

2 Exploratory Data Analysis

Before writing a single line of model code, we invested substantial effort in understanding the dataset. The full Exploratory Data Analysis report with all corresponding visualisations was submitted separately. The insights from this phase directly shaped every subsequent technical decision. Many properties of the data that we uncovered would have silently degraded our models.

2.1 Dataset Composition and Integrity

We began by building a complete dataset manifest. This is a structured table listing every image and its corresponding metadata, including product category, data split, defect type, pixel dimensions, and the path to its ground-truth mask. This manifest served as the foundation for all subsequent analyses and allowed us to run systematic integrity checks before touching any model code.

We verified that every defective test image is paired with exactly one ground-truth mask, that all image-mask dimension pairs match, and that all masks are stored in greyscale format with binary-valued pixels. We used SHA-256 hashing to check for byte-for-byte duplicates and confirmed that no exact duplicate images exist and that there is no image-level train-to-test leakage in the dataset.

In total, the dataset contains 3,629 normal training images, 467 normal test images, and 1,258 defective test images spread across 15 product categories. The training split consists exclusively of defect-free images, reflecting the unsupervised detection paradigm. We found that image modes are heterogeneous across categories: most categories contain RGB images, while the categories *grid*, *screw*, *tile*, *toothbrush*, and *zipper* use greyscale images. Image resolutions range from 700×700 to 1024×1024 pixels, and all images are square. This heterogeneity in colour mode and spatial resolution made explicit format handling and a standardised loading pipeline a necessity, as we describe in Section 4.

2.2 Objects versus Textures: A Fundamental Structural Split

The 15 categories divide into two structurally distinct groups that require entirely different treatment throughout the pipeline. The five texture categories (Carpet, Grid, Leather, Tile, and Wood) consist of repetitive, spatially invariant surface patterns. A patch of leather looks statistically the same regardless of whether it has been rotated or mirrored. The ten object categories (bottle, cable, capsule, hazelnut, metal_nut, pill, screw, toothbrush, transistor, and zipper) are physically shaped items that arrive at the inspection stage in a consistent orientation. Applying aggressive rotational augmentation to a bottle would incorrectly instruct the model that inverted orientations are part of the normal distribution, rendering the detector blind to orientation-based anomalies.

We identified this distinction on the first day of the project and carried its implications through to augmentation, foreground extraction, and configuration tuning.

2.3 Pixel-Level Class Imbalance and the Accuracy Paradox

We computed pixel-level defect statistics across all 1,258 defective test images. The median defect occupies only 1.54% of the image area. The 25th percentile is 0.49%, meaning that one quarter of all defects cover less than half a percent of the image. Some defects are as small as 0.037% of the image area, which amounts to a barely visible scratch or pinhole. The mean of

4.39% is inflated by a small number of large structural defects that reach up to 49.32% of the image.

These numbers carry a direct consequence for evaluation. A model that classifies every pixel in every image as normal achieves over 98% pixel-level accuracy on a typical defective image, while detecting exactly zero defects. Standard accuracy is therefore a completely misleading metric for this problem, which is what the literature refers to as the accuracy paradox. We concluded from this analysis that we needed to use metrics that are sensitive to the minority positive class and that do not get inflated by the large pool of correctly-ignored normal pixels. Precision-recall curves and the F1 score have exactly this property, since they are computed entirely from true positives, false positives, and false negatives. The pool of true negatives (normal pixels) never enters their calculation.

2.4 Statistical Proof of Category Heterogeneity

A recurring question during the EDA was whether a single binarisation threshold could be applied uniformly across all 15 categories. We answered this question with two statistical tests.

We first ran the Kruskal-Wallis H-Test, which tests the null hypothesis that the population medians of image-level defect area ratios are equal across all 15 categories. We chose the Kruskal-Wallis test over a pixel-level Chi-Square test for methodological reasons: pixels are strongly spatially correlated, which violates the independence assumption required by Chi-Square. Furthermore, the massive pixel sample size ($N > 67$ million) would cause the Chi-Square test to flag even negligibly small differences as statistically significant, an example of inflated statistical power overwhelming practical significance. The Kruskal-Wallis test treats each defective test image as a single independent observation ($N = 1,258$), making the inference trustworthy. The result was $H = 626.5$ with $p = 1.2 \times 10^{-124}$, a decisive rejection of the null hypothesis. Defect size distributions differ significantly across categories.

As a complementary test, we ran a pixel-level Chi-Square test to assess whether the overall proportion of anomalous pixels differs across categories. The test yielded $\chi^2(14) \approx 41.7$ million with $p \approx 0$. The anomalous-pixel ratios range from 0.34% for *screw* to 14.5% for *metal_nut*, which is a 43-fold spread.

The combined result is unambiguous: a single detection threshold applied uniformly across all 15 categories is statistically indefensible. This finding motivated our decision to tune preprocessing and model hyperparameters separately per category using Optuna.

2.5 Defect Shape, Morphology, and Spatial Location

We extended the defect analysis beyond area to include shape complexity and spatial location. For each ground-truth mask, we computed the bounding-box fill ratio, which is the fraction of the bounding box occupied by the defect, and the number of connected components, which indicates whether the defect is a single region or multiple disconnected fragments.

We found that defects vary widely in all of these dimensions. Some defects are extremely thin relative to their bounding box, for example hairline cracks in the *capsule* category. Others consist of multiple disconnected fragments, as is common in the *cable* category where damage occurs at several points simultaneously. Several categories contain defects that touch the image border, which means that any preprocessing step that crops or pads the image boundary could destroy part of the ground-truth annotation.

We also mapped the normalised spatial centre-of-mass coordinates of each defect. Rigid-object categories, particularly *bottle* and *screw*, show strong positional clustering of defects, reflecting the physical reality that certain component surfaces are more susceptible to certain types of damage. Texture categories, by contrast, show broadly distributed defect locations. We noted this positional bias as a potential shortcut for a model to exploit. A model that learns to flag the top of a bottle rather than flagging structurally anomalous regions would score well on the benchmark but fail completely in real deployment. Evaluation and augmentation decisions must be designed to prevent this.

2.6 Normal Train/Test Covariate Shift

We compared brightness and contrast distributions between normal training images and normal test images within each category. Since both sets are defect-free, any systematic difference in these statistics represents a domain shift caused by acquisition conditions rather than genuine anomalies. A model that learns the specific lighting conditions of the training set may flag normal test images as anomalous simply because the illumination differs.

We found noticeable differences in brightness or contrast for the *screw*, *leather*, and *grid* categories. This finding had two consequences: it reinforced the need for per-category evaluation rather than global averaging, and it motivated the inclusion of photometric augmentation (brightness and contrast variation) during training to make the model robust to acquisition-level variation.

2.7 Why We Do Not Use Pixel-Level AUROC

The standard metric for anomaly map quality in the literature is the pixel-level AUROC. We examined this metric carefully during the EDA and concluded that it is unsuitable as a primary evaluation standard for MVTec AD, for the same reason that accuracy is unsuitable. AUROC is built on the false positive rate, defined as $\mathrm{FPR} = \mathrm{FP}/(\mathrm{FP} + \mathrm{TN})$. Since true negatives (normal pixels) constitute approximately 98.5% of all pixels in a typical defective test image, a model can misclassify substantial numbers of pixels while the FPR barely moves. This keeps AUROC deceptively high even for weak detectors.

Published state-of-the-art results confirm this: PatchCore reports pixel AUROC of 99.6%, EfficientAD up to 99.8%, and Dinomaly 99.6%. When top systems differ by only a few tenths of a percent, AUROC cannot distinguish model quality. We therefore use pixel-level AUROC only as an internal proxy during Optuna optimisation trials where computational speed matters, and we report AUPIMO as the primary localisation metric in all final evaluations.

2.8 Threshold Sensitivity and the Business Case

As a final EDA step, we analysed the breakpoint phenomenon in threshold selection. Every anomaly detection model outputs a continuous anomaly score per pixel, and deployment requires a binary decision threshold t that separates normal pixels from anomalous ones. Selecting this threshold is not a purely statistical decision, it is a business decision with direct financial consequences.

A threshold set too high produces high precision, meaning there are few false alarms and the production line is not stopped unnecessarily, but low recall, meaning real defects escape undetected and reach customers. A threshold set too low catches more defects and yields high recall, but triggers frequent unnecessary line stoppages, which leads to high cost from lost throughput. We computed the F1 score, which balances precision and recall, the F2 score,

which places twice the weight on recall for applications where missed defects are unacceptably costly, and the F0.5 score, which places twice the weight on precision for applications where false alarms dominate costs. This allowed us to map the full trade-off quantitatively. The threshold sensitivity analysis showed that precision and recall react sharply to small threshold changes, making the choice of operating point non-trivial even within a single category.

3 Evaluation Metrics

Based on the findings of the EDA, we selected a set of evaluation metrics that operate at two distinct levels, image-level and pixel-level, and that are honest under the extreme class imbalance of the dataset.

3.1 Image-Level: F1 Score and Precision-Recall Curve

At image level, the task is binary classification. Given a test image, we want to know whether it is normal or defective. We compute the F1 score, defined as the harmonic mean of Precision P and Recall R:

$$F_1 = 2 \cdot \frac{P \cdot R}{P + R}, \quad \text{where} \quad P = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad \text{and} \quad R = \frac{\text{TP}}{\text{TP} + \text{FN}}.$$

Here, a true positive (TP) is a defective image that the model correctly flags, a false positive (FP) is a normal image that the model incorrectly flags, and a false negative (FN) is a defective image that the model misses. The denominator of precision contains only images that the model actually flagged. It does not involve the large pool of correctly-accepted normal images. This makes F1 immune to the accuracy paradox. The image-level classification threshold is derived exclusively from normal validation images, as described in Section 7.3.

We complement the F1 score with the full precision-recall curve, which traces the trade-off across all possible threshold values. The area under this curve summarises ranking quality in a single number that is fully sensitive to the imbalance between normal and defective images.

3.2 Pixel-Level: AUPIMO

At pixel level, the task is to produce an anomaly map $a \in \mathbb{R}^{H \times W}$ such that anomalous pixels receive high scores and normal pixels receive low scores. We use AUPIMO [5] as the primary localisation metric.

AUPIMO is built on two core ideas. The first is normal-only threshold calibration. For a given threshold t, the shared false positive rate $F_{sh}(t)$ is computed exclusively over the set of strictly normal images $\mathcal{Y}^0$:

$$F_{sh}(t) = \frac{1}{|\mathcal{Y}^0|} \sum_{y \in \mathcal{Y}^0} \frac{|\{(i,j) : a_{ij}^{(y)} \geq t\}|}{H \cdot W}.$$

The threshold calibration is completely blind to anomalous test images. It cannot access ground-truth defect locations, and there is no risk of the threshold selection leaking information about where defects are.

The second idea is per-image scoring. Rather than pooling all pixels from all test images into a single global FPR-TPR trade-off, AUPIMO computes the true positive rate $T_i(t)$ independently for each anomalous image $y_i \in \mathcal{Y}^1$:

$$T_i(t) = \frac{|\{(j,k) : a_{jk}^{(y_i)} \geq t\} \cap \{(j,k) : m_{jk}^{(y_i)} = 1\}|}{|\{(j,k) : m_{jk}^{(y_i)} = 1\}|},$$

where $m^{(y_i)}$ is the binary ground-truth mask of image y_i. Every image contributes equally to the final score regardless of its defect size.

The final AUPIMO score is computed as the integral of T_i over a logarithmically-spaced range of Shared FPR values:

$$\text{AUPIMO}_i = \frac{1}{\ln(U/L)} \int_L^U T_i\big(F_{sh}^{-1}(z)\big) \frac{\mathrm{d}z}{z}, \quad L = 10^{-5},\ U = 10^{-4}.$$

The integration bounds $L = 10^{-5}$ and $U = 10^{-4}$ are chosen to reflect the operating regime of real industrial deployment. For a 256×256 image containing 65,536 pixels, $F_{sh} = 10^{-5}$ corresponds to fewer than one false positive pixel per image on average. Evaluating in this regime is what distinguishes AUPIMO from AUROC, because state-of-the-art models that score 99% under pixel AUROC typically score 60-70% under AUPIMO, revealing that current benchmarks are far from solved at industrially meaningful false positive rates.

3.3 Threshold Sensitivity Curves

For each model, we also report threshold sensitivity curves that show how F1 and AUPIMO evolve as the classification threshold t varies. These curves reveal whether a model's performance is stable across a range of thresholds or collapses abruptly at a specific threshold value. A robust threshold curve is preferable in deployment, because real production conditions always involve some drift in image characteristics over time.

4 Preprocessing Pipeline

All images pass through a standardised preprocessing pipeline before entering any model. We designed this pipeline in direct response to the EDA findings concerning the heterogeneity of image formats, the object-versus-texture distinction, the risk of covariate shift from background variation, and the fragility of small defects to aggressive resizing.

4.1 Loading, Format Standardisation, and Resizing

We load every image using PIL and immediately convert it to three-channel RGB. This step is mandatory because MVTec images exist in four distinct formats: RGB JPEG, RGBA PNG, 8-bit greyscale PNG, and palette-mapped PNG. Without explicit conversion, feeding a 1-channel greyscale image to a network expecting 3 channels would produce a shape mismatch.

We resize all images to 256×256 pixels using LANCZOS resampling. LANCZOS implements a windowed sinc function (mathematically defined as $\text{sinc}(x) = \frac{\sin(x)}{x}$), which acts as an ideal mathematical low-pass filter. This mathematically minimises aliasing artefacts while preserving edge sharpness and high-frequency texture detail. We chose it specifically over bilinear or bicubic

resampling because some defects are microscopic, and a lower-quality interpolation filter could destroy that signal entirely during downscaling.

Ground-truth defect masks are resized separately using nearest-neighbour interpolation. We deliberately do not use anti-aliasing filters for masks because anti-aliasing would blur the binary mask boundaries, introducing fractional pixel values at the edges of annotated defect regions. After resizing, we binarise all masks at the threshold 127 to ensure strict binary arrays.

4.2 Category-Aware Augmentation

We apply data augmentation to the training set only, in order to increase the effective diversity of the training distribution. The augmentation strategy differs fundamentally between texture and object categories, as motivated by the EDA.

For texture categories, we apply heavy spatial augmentation. We randomly rotate images in 90 degree increments, flip them horizontally and vertically, and apply scale jitter by cropping a region sized between 80% and 100% of the original image and resizing it back to the target resolution. We additionally vary brightness and contrast by up to 20%. Every spatially augmented version of a normal texture image is a genuinely valid training example, because texture surfaces are statistically indistinguishable at arbitrary orientations.

For object categories, we apply photometric augmentation only. We vary brightness, contrast, and saturation each by up to 10% and add Gaussian noise with standard deviation $\sigma = 0.02$. We do not apply any spatial transformation, because an object's orientation on the inspection conveyor is part of its normal appearance, and training on arbitrarily oriented objects would destroy the model's ability to detect orientation-based anomalies.

4.3 Foreground Extraction

Industrial inspection images always include a background such as a conveyor belt surface or a mounting jig. This background wastes model capacity and creates a source of false positives. Any slight variation in the background, like a shadow or a piece of dust, generates a reconstruction error that the model may misidentify as a product defect.

We remove backgrounds using a two-stage classical computer vision pipeline. In the first stage, we apply Otsu's global thresholding method [10] to the greyscale version of the image. Otsu's method finds the threshold T^* that maximises the between-class variance:

$$\sigma_B^2(T) = w_0(T) \cdot w_1(T) \cdot [\mu_0(T) - \mu_1(T)]^2 ,$$

where $w_0(T)$ and $w_1(T)$ are the fractions of pixels assigned to the background class and foreground class respectively, and $\mu_0(T)$ and $\mu_1(T)$ are their corresponding mean intensities. This method works automatically because it adapts to the pixel intensity histogram of each individual image.

In the second stage, we apply Canny edge detection [11] to find the sharp object boundaries that Otsu's region-growing approach may miss. Canny operates using Gaussian smoothing, Sobel gradient computation, non-maximum suppression, and hysteresis thresholding. We set the two Canny thresholds adaptively at $\pm 33\%$ around the image's median pixel intensity, making the detector self-calibrating across varied lighting conditions.

The binary masks produced by Otsu and Canny are merged with a logical OR. We then apply a 5×5 morphological closing operation to fill any remaining holes. Finally, we retain only the

largest connected component and discard all others, removing dust particles and small image artefacts. The background is replaced with pure black, ensuring that the model produces zero reconstruction error in background regions.

4.4 Optional Enhancement Steps

The pipeline exposes two additional preprocessing steps that are not applied globally but are toggled per category by the Optuna hyperparameter sweep (Section 8).

The Contrast Limited Adaptive Histogram Equalization (CLAHE) divides the image into a regular grid of 8×8 pixel tiles and applies local histogram equalisation within each tile, with a clip limit of 2.0 that prevents over-amplification of noise in uniform regions. CLAHE enhances subtle surface textures, faint stains, and low-contrast scratches. The associated risk is that CLAHE also amplifies normal microscopic surface variations, which may cause a sensitive model to flag them as structural anomalies.

The Gaussian Blur step applies a 5×5 Gaussian kernel to the image before it enters the model. The blur acts as a low-pass filter that suppresses high-frequency sensor noise and microscopic surface irregularities. The risk is that blurring destroys fine-grained detail, and if the defects being targeted are microscopic, the blur may erase them from the image before the model can detect them.

We observed during initial experiments that applying both CLAHE and Gaussian Blur together can introduce visual artifacts. CLAHE amplifies high-frequency detail, and Gaussian Blur then smears it into a cloud of intermediate-frequency patterns that neither the normal nor the defective training distribution resembles. We therefore let the Optuna sweep determine the combination of these steps that maximises pixel AUROC for each category individually.

5 Exploratory Autoencoder Development

Our first model development phase consisted of iterative notebook-based experiments. We focused exclusively on the *bottle* category to maintain a consistent comparison base. Under the intended deterministic 85%/15% protocol, its 209 normal development images are divided into 177 training images and 32 validation images; the official test set contains 83 images and remains separate. We deliberately chose to document this process as a progression of insights, because the dead ends and missteps are as informative as the improvements.

We developed several autoencoder variants over the course of the exploration phase. We present four representative milestone architectures that capture the key turning points in our understanding, followed by the final production model as a fifth architecture. We made the decision not to discuss every single iteration to avoid obscuring the main conceptual arc of the project.

5.1 The Reconstruction Principle

All autoencoder experiments share the same detection principle. An autoencoder consists of an encoder that compresses the input image into a compact latent representation, and a decoder that reconstructs the image from the latent code. During training, both networks are jointly optimised to minimise the reconstruction loss on normal images only.

At inference time, the pixel-level reconstruction error serves as the anomaly map. A well-trained autoencoder produces low error on normal images and high error at defective regions, because it

has never learned how anomalous structures look and reconstructs the expected normal surface instead.

The image-level anomaly score aggregates the per-pixel error into a single scalar value. To compute the classification threshold, we collect these scalar anomaly scores for all normal validation images, sort them in ascending order, and select the value below which 95% of the scores fall. This 95th percentile then serves as our classification threshold.

5.2 Milestone 1: Three-Stage MSE Autoencoder

Our first architecture, Milestone 1, used a symmetric three-stage convolutional encoder-decoder. The encoder applied three stride-2 convolutions with a kernel size of 4×4, progressively doubling the number of feature channels from 32 to 128 while halving the spatial dimensions from 256×256 to 32×32. The decoder mirrored this structure with three transposed-convolution stages. The bottleneck held 131,072 values, giving a compression ratio of approximately 1.5 relative to the input. We used ReLU and LeakyReLU activations during this early phase, which we would later replace with ELU.

Input	→	Enc-1	→	Enc-2	→	Bottleneck
256×256×3		Conv(32)		Conv(64)		128×32×32

Bottleneck	→	Dec-2	→	Dec-1	→	Output
128×32×32		ConvT(64)		ConvT(32)		256×256×3

We trained with mean squared error (MSE) loss and the Adam optimiser. We extended the training progressively because validation loss was still decreasing at the end of each run, until we achieved the lowest validation reconstruction MSE of the group at 200 epochs.

On the bottle test set, this model produced an image-level AUROC of 0.617 and detected 24 of 63 defects. These are weak results. More strikingly, the version with the best reconstruction quality on normal images was the worst anomaly detector of the entire exploration phase.

From these initial experiments, we discovered the core failure mode of autoencoders trained purely on mean squared error. A low reconstruction error on normal images does not imply a correspondingly high reconstruction error on defective images. A model trained to minimise the mean squared error develops a high capacity to reconstruct any smooth image, including images with defects. Consequently, the reconstruction error becomes uninformative as an anomaly signal. The correct goal is not to achieve the lowest possible reconstruction error on normal images, but rather to maximise the difference in reconstruction error between normal and defective images.

5.3 Milestone 2: First Masked Restoration Experiment

The identity-mapping problem, where the autoencoder learns to copy its input rather than learning the structure of normal images, suggested that the training task itself needed to change. We introduced our first masking strategy in Milestone 2, keeping the same three-stage architecture but replacing the clean reconstruction objective with a masked restoration task. During training, we randomly selected two non-overlapping 32×32 patches in the input image and replaced them with the mean colour of the image. The model could no longer copy its input, as it had to use the surrounding context to infer what should appear in the masked areas.

We evaluated the trained checkpoint using a clean-input scoring protocol where we fed the original unmasked image through the encoder, and a masked-restoration scoring protocol where we tiled the image with a fixed set of deterministic masks.

Scoring method	AUROC	PR-AUC	Bal. Acc.	Defects found
M1 (clean-input)	0.617	0.860	0.641	24/63
M2 (clean-input)	0.763	0.924	0.766	43/63
M2 (masked-restoration)	0.684	0.900	0.713	30/63

The masked training objective improved clean-input scoring substantially, but the masked-restoration scoring protocol did not outperform clean-input scoring and missed more defects.

This milestone demonstrated that masked training changes what the model learns, but it does not automatically produce better anomaly scores. The scoring protocol is a separate design choice from the training task. An effective training objective paired with a poorly matched scoring method still yields weak results.

5.4 Milestone 3: Stronger Spatial Compression

Milestone 3 introduced a fourth encoder stage that reduces the latent spatial resolution from 32×32 to 16×16. The bottleneck became $128 \times 16 \times 16 = 32,768$ values, which is a compression factor of 6 relative to the input. The encoder had four stride-2 convolutional stages, and the decoder mirrored it with four transposed-convolution stages.

In	→	E-1	→	E-2	→	E-3	→	Bottleneck
$256^2 \times 3$		Conv(32)		Conv(64)		Conv(128)		$128 \times 16 \times 16$

Bottleneck	→	D-3	→	D-2	→	D-1	→	Out
$128 \times 16 \times 16$		ConvT(128)		ConvT(64)		ConvT(32)		$256^2 \times 3$

We kept the MSE loss, clean-input scoring, and the same 200-epoch training schedule. The result was an AUROC of 0.877 and 49 of 63 defects detected. This was the best autoencoder result of the exploration phase by a wide margin.

These results confirmed that stronger compression is the correct architectural lever for reconstruction-based anomaly detection. By squeezing the representation into a 16×16 spatial grid, we forced the model to discard fine-grained local texture detail and retain only the coarser structural grammar of normal bottles. Anomalous structures cannot be faithfully reconstructed from this highly compressed representation, which in turn produces a visible reconstruction error.

We also tested extending the training to 400 epochs, and narrowing the bottleneck further to 32 channels. Both variants performed worse than Milestone 3. Longer training caused the model to over-generalise, and excessive bottleneck narrowing prevented faithful reconstruction even of normal images.

5.5 Milestone 4: Combined Masking and Compression

Milestone 4 combined the masked restoration training objective of Milestone 2 with the deeper four-stage architecture of Milestone 3. We kept the $128 \times 16 \times 16$ bottleneck and the masked-patch loss, and trained for 150 epochs.

Masked-restoration scoring on Milestone 4 achieved an AUROC of 0.822, a clear improvement from combining stronger compression with the masking task. However, it still did not match Milestone 3's clean-input result of 0.877. The two ideas interacted positively but produced a combined result that fell between their individual bests.

This outcome highlighted that combining two individually beneficial modifications does not guarantee a result that surpasses either one alone. The experiment validated that masking

helps more when combined with stronger compression, but the masking strategy still required further refinement before it could outperform the pure compression approach. This insight directly motivated the design of our final Keras autoencoder.

5.6 The Pretrained Baseline: ResNet-18 + 5-NN

During the exploratory phase, it became evident that training an autoencoder from scratch on a highly constrained normal-only dataset is challenging. To properly contextualise the autoencoder results, we established a simple comparative baseline using transfer learning. We extracted global features from a frozen ResNet-18 network pretrained on ImageNet and applied a basic 5-Nearest Neighbour (5-NN) scoring algorithm.

This simple pretrained baseline achieved an AUROC of 0.9881, a PR-AUC of 0.9960, an F1 score of 0.9594, and a balanced accuracy of 0.9433 on the *bottle* test set. At its normal-validation threshold, it detected 59 of 63 defects while raising only one false alarm among 20 normal test images.

This result starkly contrasted with our best autoencoder configuration at the time (AUROC 0.877, 49/63 defects). It provided the decisive proof that frozen ImageNet features dramatically outperform models trained from scratch for this specific task, directly motivating our subsequent adoption of the full PatchCore architecture (Section 7).

6 The Final Keras CAE (Milestone 5)

Based on the four milestones, we designed the final production autoencoder in Keras. This model incorporates the main successful ideas while addressing the remaining weaknesses identified during the exploration phase. We structured the Keras code using a sequential model setup to ensure clean layer definitions and manageable state.

6.1 Architecture and Activation Layers

The encoder uses four stride-2 convolutional stages with 4×4 kernels and a channel progression of $3 \to 32 \to 64 \to 128 \to 256$. The decoder mirrors the encoder with four transposed-convolution stages of matching channel widths. A final 1×1 convolutional layer reduces the bottleneck from 256 to 32 channels while preserving the 8×8 spatial grid. This fully convolutional bottleneck contains $32 \times 8 \times 8 = 2,048$ values, operating on 128×128 input images to give a compression ratio of approximately 24.

In		E-1		E-2		E-3		E-4		Bottleneck
128^2×3	→	Conv(32) BN+ELU	→	Conv(64) BN+ELU	→	Conv(128) BN+ELU	→	Conv(256) BN+ELU	→	8×8×32 FCAE

Bottleneck		D-4		D-3		D-2		D-1		Out
8×8×32 FCAE	→	ConvT(256) BN+ELU	→	ConvT(128) BN+ELU	→	ConvT(64) BN+ELU	→	ConvT(32) Sigmoid	→	128^2×3

By calculating the spatial dimensions and channel depths, we can trace the total number of activations (or "neurons') passed through each stage of the network. The 128×128 RGB input contains 49,152 values. As the spatial dimensions halve and channels double with each stride-2 convolution, the encoder activations progress from 131,072 (E-1) → 65,536 (E-2) → 32,768 (E-3) → 16,384 (E-4), finally condensing into just 2,048 neurons in the $8 \times 8 \times 32$ bottleneck. The

decoder then expands this compressed representation, peaking at 524,288 neurons in D-1 before collapsing back to the 49,152 output pixels.

We moved away from the early ReLU and LeakyReLU activations and used Exponential Linear Unit (ELU) activations throughout the network. The ReLU sets its output to zero for all negative inputs, which can cause neurons that consistently receive negative activations to produce zero gradient permanently. ELU avoids this by saturating smoothly toward a negative value for negative inputs, ensuring that the gradient always flows. We use Batch Normalisation after every convolution, which re-centres and re-scales activations within each mini-batch. All convolutional layers omit the bias term, because the Batch Normalisation layer's learnable shift parameter subsumes any constant bias.

The bottleneck is fully convolutional. Rather than flattening the feature maps into a 1D vector, we apply an additional 1×1 convolution that reduces the channel count while retaining the 8×8 spatial grid. During our manual testing on the bottle dataset, we actually tried using a central flatten layer followed by a dense layer for the bottleneck. We observed that this produced a good binary classification score for whether the image as a whole was anomalous or not, but it completely scrambled the spatial localisation map. The decoder had to re-learn coordinate relationships from the flattened vector from scratch. By preserving the spatial topology in the fully convolutional bottleneck, we allowed the decoder to maintain and refine spatial relationships during reconstruction.

6.2 Masked Image Modelling and Randomisation

We replaced the coarse two-patch masking of Milestone 2 with a principled Masked Image Modelling strategy, implemented as a custom randomisation layer in the sequential Keras pipeline. The layer divides the image into a regular grid of non-overlapping 16×16 patches. At each training step, the randomisation layer randomly selects 25% of these patches and sets their pixel values to zero. The encoder receives the masked image, and the training target is the original unmasked image.

The random patch selection is re-drawn independently for every image and every training step, ensuring that no patch position is consistently masked. At inference time, the randomisation layer is disabled, and the test image is passed through the network unmodified.

6.3 Combined SSIM and MSE Loss

We replaced the pure MSE training loss with a combination of the structural similarity index measure (SSIM) and MSE. The combined loss is defined as:

$$\mathcal{L}(x, \hat{x}) = \alpha \cdot (1 - \mathrm{SSIM}(x, \hat{x})) + (1 - \alpha) \cdot \mathrm{MSE}(x, \hat{x}), \quad \alpha = 0.84.$$

SSIM evaluates image quality through three complementary local statistics computed over a sliding window. Because SSIM compares local structural patterns rather than individual pixel values, it is robust to the kind of sub-pixel spatial shifts that cause pure pixel-wise losses to be uniformly elevated.

We note that the original publication by Bergmann et al. [3] evaluated SSIM in combination with the L1 loss (Mean Absolute Error) and L2 loss (MSE) and recommended L1 for their specific configuration. We implemented MSE in combination with SSIM because it provided excellent convergence in our Keras pipeline, covering absolute photometric errors while SSIM

handles structural differences. The weight parameter $\alpha = 0.84$ controls the balance between structural fidelity and per-pixel accuracy.

6.4 Optimiser and Callbacks

We moved from the standard Adam optimiser to the AdamW optimiser [7]. Standard Adam incorporates weight decay by adding it to the gradient before the gradient enters the adaptive moment estimates. This causes weights with large gradient histories to be under-regularised. AdamW decouples the weight decay and applies it as a direct multiplicative shrinkage to each weight independently, ensuring that every weight receives exactly proportional regularisation.

The training process is governed by a set of Keras callbacks that monitor the validation reconstruction loss. We use the *EarlyStopping* callback with a patience of 20 epochs to halt training when the model stops improving, preventing over-generalisation. This is paired with the *ReduceLROnPlateau* callback, which halves the learning rate when the validation loss has not improved for 5 consecutive epochs, allowing the optimiser to settle into narrow local minima. Finally, the *ModelCheckpoint* callback ensures that the network weights from the epoch with the lowest validation loss are restored as the final model, regardless of when the early stopping condition was triggered.

7 PatchCore Baseline

In parallel with the autoencoder experiments, we established a state-of-the-art baseline using PatchCore [4]. PatchCore requires no gradient-based training, and we use it both as a performance upper bound for the autoencoder and as a standalone production system.

7.1 Method Description

PatchCore operates by passing all normal training images through a frozen pretrained convolutional network. The extracted feature maps are spatially aggregated by computing a neighbourhood average for each spatial position. The collection of all these feature vectors forms a memory bank that represents the full distribution of normal patch appearances.

To manage the memory footprint, PatchCore applies a greedy coreset subsampling step that reduces the memory bank to a configurable fraction of its original size while maintaining good geometric coverage of the distribution. At inference time, each patch feature extracted from a test image is scored by its Euclidean distance to the nearest neighbour in the memory bank. A patch far from all normal patches receives a high anomaly score.

7.2 Backbone: ResNet-18

We use ResNet-18 pretrained on ImageNet as the PatchCore backbone. The larger Wide-ResNet-50-2 backbone triggered repeated out-of-memory exceptions during automated Optuna sweeps. These failures caused the Optuna study to miss entire hyperparameter configurations. We deprecated Wide-ResNet-50-2 and hardcoded ResNet-18 for all sweep runs to ensure stable, reproducible execution.

7.3 Data Split and Threshold Calibration

For every category, we deterministically divide the official normal training partition into 85% model-training data and 15% normal validation data using random seed 42. The CAE learns its

weights from the 85% subset, while PatchCore builds its feature memory bank and coreset from the same 85% subset. The 15% subset is held out from both models. It is used for CAE early stopping and for calibrating each model's image-level operating threshold. We set that threshold from the distribution of normal validation scores and then freeze it before test inference. The official test images, labels, and masks are used only for final metrics and qualitative overlays.

AUPIMO requires a careful distinction. It is not the deployed image-level operating threshold. During final evaluation, AUPIMO uses normal test anomaly maps to construct its shared false-positive-rate axis and anomalous test masks to calculate per-image recall. This use of the test set is intrinsic to computing the final metric and does not select model settings. The FPR bounds $(10^{-5}, 10^{-4})$ and numerical threshold resolution are fixed before the test run. No AUPIMO result may be fed back into training or model choice during development.

The evaluation setup is therefore as follows:

1. **Identical data partitions.** Both models use the same exact image lists: 85% of the official normal training images for fitting and 15% for validation and calibration. The split is generated once with random seed 42 and reused by both pipelines. For *bottle*, this gives 177 fitting images and 32 validation images.
2. **Validation-derived image thresholds.** Each model produces image-level anomaly scores for the same 15% validation images. Its deployed operating threshold is calculated only from those normal validation scores and is frozen before the test set is opened. Normal test scores must not determine this threshold.
3. **Identical final test set and reporting.** Both models are evaluated on the same ordered test-image list. For *bottle*, this is the complete set of 83 official test images. We report the confusion counts TP, FP, and FN together with precision, recall, and F1, rather than reporting F1 alone.
4. **Canonical pixel resolution.** Before pixel metrics are compared, both models' anomaly maps and ground-truth masks are converted to 256×256. Continuous anomaly maps are resized with bilinear interpolation, while binary masks use nearest-neighbour interpolation and remain binary.
5. **Identical metric implementations.** Pixel AUROC is computed through one shared function for both models. AUPIMO is computed through the same full-map `anomalib.metrics.AUPIMO` path using complete two-dimensional anomaly maps and masks, fixed FPR bounds $(10^{-5}, 10^{-4})$, and `num_thresholds` set to 50,000. Alternative FPR bounds, pooled-pixel substitutes, and fallback metric values are not permitted; a failed metric calculation must be reported as an error.
6. **Separation of threshold roles.** The image-classification operating threshold comes from the 15% validation set. AUPIMO's shared FPR axis is instead calculated from normal images in the final test/evaluation set because this is part of AUPIMO's metric definition. These operations serve different purposes and must not be conflated.

8 Hyperparameter Optimisation with Optuna

The EDA showed substantial differences between object and texture categories, so we investigated whether category-specific hyperparameter optimisation could improve both model families. Given the size of the search space, manual experimentation or exhaustive grid search was im-

practical. We therefore used Optuna [12] as an experimental branch of the project rather than assuming that one configuration would suit every category.

8.1 How Optuna Works (The TPE Algorithm)

Optuna uses a Tree-structured Parzen Estimator (TPE), a form of Bayesian optimisation. TPE learns from completed trials by modelling promising and unpromising regions of the search space, then prioritises configurations that are more likely to improve the objective. This made it suitable for testing several interacting architectural and preprocessing choices within our limited compute budget.

8.2 Search Spaces for PatchCore and Keras CAE

For PatchCore, we explored which ResNet-18 feature layers contributed to the memory bank, coreset sampling ratios between 0.001 and 0.2, nearest-neighbour values from 1 to 15, and optional CLAHE, Gaussian blur, and foreground masking. For the Keras CAE, we explored bottleneck dimensions from 16 to 128 channels together with the same optional preprocessing choices. The studies were configured for 30 trials per category for each model family across six representative categories—*bottle*, *cable*, *capsule*, *carpet*, *grid*, and *hazelnut*—giving up to 360 planned trials, although hardware failures interrupted some runs.

The exploratory objectives differed by model: PatchCore maximised pixel-level AUROC, while the Keras CAE maximised pixel-level AUPIMO. These early searches obtained objective feedback from the official test partition. We therefore treat them only as evidence of our experimental process, not as valid final results, and exclude all tuned metrics from the fair baseline comparison.

8.3 Practical Difficulties During the Sweep Process

The tuning process produced several useful engineering lessons. Wide-ResNet-50-2 repeatedly exceeded the available GPU memory during PatchCore trials, which led us to standardise on ResNet-18. Sequential TensorFlow trials also accumulated memory, so we designed an orchestrator that ran each CAE trial in an isolated subprocess and allowed the operating system to reclaim its resources afterwards.

More importantly, additional optimisation did not translate into reliable improvements. Tuned configurations frequently performed worse than the simpler baselines, and optional preprocessing could suppress subtle defect evidence while improving the chosen proxy objective. The experiment showed that a more extensively tuned model is not necessarily a better anomaly detector, particularly when optimisation and final evaluation do not use an identical leakage-free protocol. Given the computational cost, inconsistent gains, and methodological limitations of the exploratory objectives, we rejected the tuned variants. The final comparison therefore evaluates only the two reproducible baseline pipelines across all 15 categories.

9 Evaluation Results

In this section, we present the completed evaluation of the PatchCore and Keras CAE baselines. We first describe the shared evaluation procedure, then examine anomaly-map quality, quantitative results, and precision-recall and threshold curves.

9.1 Evaluation Procedure

We split the official normal MVTec training partition into 85% for model fitting and 15% for validation for both models, using random seed 42. The CAE trains its weights on the 85% subset; PatchCore constructs its memory bank and coreset from that same subset. The held-out 15% is used for CAE early stopping and for normal-only operating-threshold calibration. It is not included in either model's fitted representation.

The official test set is not used for fitting, validation, threshold calibration, or model selection. It is accessed only after the model and operating threshold have been frozen, for one final evaluation. AUPIMO is computed at this stage from complete two-dimensional test anomaly maps and masks with fixed FPR bounds $(10^{-5}, 10^{-4})$; its use of normal test maps to define the shared FPR axis is part of the metric calculation itself, not a development decision.

The baseline PatchCore and Keras CAE models were evaluated across all 15 MVTec categories. The earlier tuned experiments are retained as part of the learning narrative in Section 8, but their metrics are not included in the final comparison.

9.2 Anomaly Map and Localisation Quality

We have placed qualitative visualisations of the anomaly maps in the Appendix to keep the main text readable. For six representative categories, the baseline PatchCore examples show the original image, ground-truth mask, anomaly-map overlay, and validation-thresholded prediction. The baseline Keras CAE examples use paired anomaly-map and ground-truth-overlay views.

When interpreting these visualisations, it is important to understand the colour scaling. The heatmaps map raw anomaly scores to a continuous colour spectrum, where dark regions (dark blue or black) indicate normal, low-score areas, and bright, warmer colours (yellows, oranges, and reds) indicate high-confidence anomalies. Consequently, a highly precise model like PatchCore often produces very dark, sparse heatmaps with a single bright peak tightly clustered around the true defect. Conversely, models that struggle with high-frequency background reconstruction (such as the Keras CAE baseline) tend to produce much brighter, noisier heatmaps overall, as their baseline anomaly score remains elevated across the entire image. In practice, this allows for a much more direct visual assessment of how effectively each model actually localises structural defects instead of just flagging background noise.

9.3 Precision-Recall Curves and Threshold Sensitivity

Looking at the precision-recall (PR) and threshold trade-off curves provides a deeper view into the models' underlying operating characteristics. We display the complete set of curves for the *bottle* category directly in Section 9.5. For all the remaining categories we evaluated, you can find the corresponding curves provided in the Appendix alongside the heatmaps.

9.4 Quantitative Results

Table 1 records the completed baseline evaluation across all 15 categories using the three metrics most relevant to the final comparison. Image-level F1 measures the frozen operating decision, image-level PR-AUC measures threshold-independent detection ranking under class imbalance, and AUPIMO measures localisation in the low-false-positive regime. Complete operational, confusion-matrix, pixel-level, and protocol metrics are reported in Appendix 10.

PatchCore achieved mean image-level F1 of 0.929, mean image-level PR-AUC of 0.988, and mean AUPIMO of 0.603. The Keras CAE achieved corresponding means of 0.495, 0.867, and

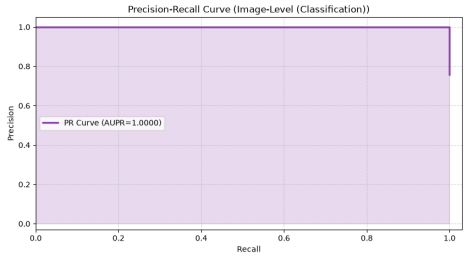
0.058. PatchCore therefore produced both substantially more reliable image-level decisions and more precise anomaly localisation. The difference between PR-AUC and F1 is also informative: a model can rank defective images reasonably well while still performing poorly at its validation-calibrated operating threshold.

Table 1: Primary baseline evaluation results across all 15 categories. PR-AUC is computed by trapezoidal integration of the stored image-level precision-recall curve.

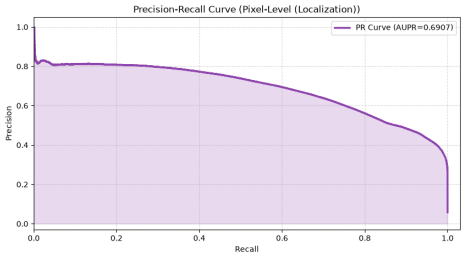
Category	**PatchCore (ResNet-18)**			**Keras CAE (Baseline)**		
	Img F1	**Img PR-AUC**	**AUPIMO**	**Img F1**	**Img PR-AUC**	**AUPIMO**
Bottle	0.962	1.000	0.982	0.992	1.000	0.507
Cable	0.933	0.985	0.511	0.154	0.624	0.000
Capsule	0.982	0.996	0.563	0.476	0.917	0.052
Carpet	0.926	0.994	0.796	0.635	0.835	0.003
Grid	0.916	0.986	0.506	0.602	0.901	0.077
Hazelnut	0.971	0.999	0.863	0.500	0.902	0.033
Leather	0.898	0.998	0.972	0.786	0.892	0.075
Metal Nut	0.963	0.998	0.760	0.243	0.825	0.002
Pill	0.891	0.982	0.293	0.395	0.923	0.030
Screw	0.796	0.983	0.380	0.421	0.887	0.005
Tile	0.951	0.993	0.685	0.132	0.727	0.001
Toothbrush	0.923	0.949	0.414	0.735	0.922	0.054
Transistor	0.916	0.976	0.427	0.657	0.758	0.005
Wood	0.952	0.995	0.642	0.182	0.955	0.008
Zipper	0.962	0.985	0.251	0.513	0.940	0.025
Mean	**0.929**	**0.988**	**0.603**	**0.495**	**0.867**	**0.058**

9.5 Evaluation Curves

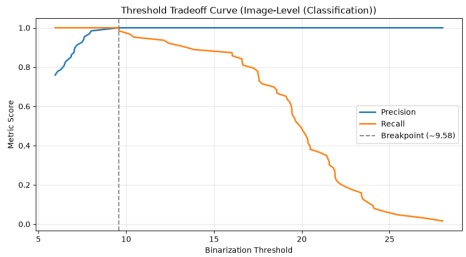
We present precision-recall and threshold trade-off curves to provide a deeper view into the models' operating characteristics. We display both baseline models for the *bottle* category below. Curves for five additional representative categories are provided in the Appendix.



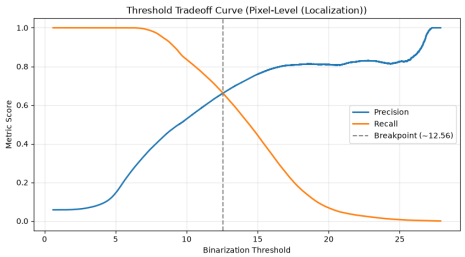
(a) Image-Level PR Curve



(b) Pixel-Level PR Curve

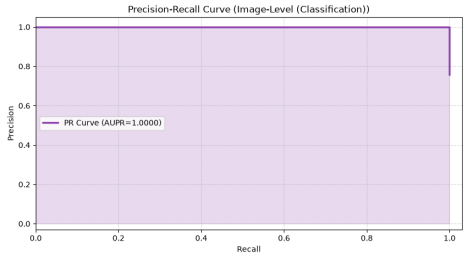


(c) Image-Level Threshold Tradeoff

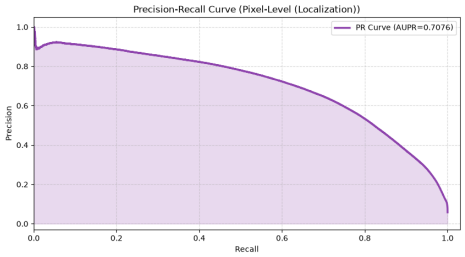


(d) Pixel-Level Threshold Tradeoff

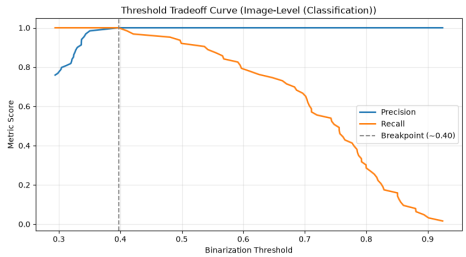
Figure 1: Evaluation curves for the *bottle* category using PatchCore (ResNet-18). The pixel-level AUPIMO is 0.982.



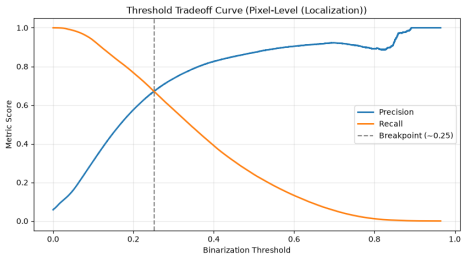
(a) Image-Level PR Curve



(b) Pixel-Level PR Curve



(c) Image-Level Threshold Tradeoff



(d) Pixel-Level Threshold Tradeoff

Figure 2: Evaluation curves for the *bottle* category using Keras CAE (Baseline). The pixel-level AUPIMO is 0.507.

9.6 Discussion of Evaluations

The completed baseline results cover all 15 MVTec categories. The Optuna experiments described in Section 8 informed our learning process but are excluded from this comparison.

1. PatchCore + ResNet-18 (Baseline): This setup establishes a strong baseline, with mean image-level F1 of 0.929 and mean AUPIMO of 0.603. It achieved image F1 above 0.89 in 14 of the 15 categories; *screw* was the most difficult at 0.796. Localisation varied more strongly by category, from AUPIMO 0.982 for *bottle* and 0.972 for *leather* to 0.251 for *zipper*. The overall performance demonstrates the advantage of pretrained ImageNet features for both detection and localisation.

2. Keras CAE (Baseline): The baseline Keras convolutional autoencoder, operating without category-specific preprocessing and with a fixed latent dimension of 32, performs strongly on *bottle*, with image F1 of 0.992 and AUPIMO of 0.507. Its performance is much less stable across the complete dataset: mean image F1 is 0.495 and mean AUPIMO is 0.058. It struggles particularly on *tile*, *cable*, and *wood*. Even where pixel AUROC is high, low AUPIMO shows that reconstruction error often remains too spatially diffuse for precise localisation at a low false-positive rate.

10 Conclusion

We developed an end-to-end unsupervised anomaly detection system for industrial components, building a production-grade modular pipeline and investigating automated hyperparameter optimisation as an experimental branch. We traced the complete arc of the project, including the conceptual missteps and rejected variants, because these were central to what we learned.

Across all 15 categories, PatchCore clearly outperformed the Keras CAE baseline. Its mean image F1 was 0.929 compared with 0.495, and its mean AUPIMO was 0.603 compared with 0.058. The CAE remained competitive on *bottle*, but its reconstruction errors were not sufficiently selective across more complex objects and textures. We therefore select PatchCore with the ResNet-18 backbone as the final model. The CAE remains valuable as an interpretable reconstruction-based reference and as evidence of the limitations of training anomaly representations from scratch on small normal-only datasets.

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Appendix: Complete Metrics and Visualisations

Complete Baseline Metrics

Tables 2–4 report the complete, non-duplicated performance metrics collected for the two final baseline models. Image PR-AUC is obtained by trapezoidal integration of each stored precision-recall curve. Accuracy is intentionally omitted because the normal/defective class imbalance makes it less informative than precision, recall, F1, and the confusion counts. Repeated aliases and anomaly-map range diagnostics are likewise omitted because they are not independent performance measures.

Table 2: Complete image-level ranking and operating-point metrics. All operating-point metrics use category-specific thresholds calibrated exclusively from normal validation images.

Category	PatchCore (ResNet-18)					Keras CAE (Baseline)				
	AUROC	PR-AUC	F1	Precision	Recall	AUROC	PR-AUC	F1	Precision	Recall
Bottle	1.000	1.000	0.962	0.926	1.000	1.000	1.000	0.992	1.000	0.984
Cable	0.975	0.985	0.933	0.955	0.913	0.467	0.624	0.154	0.667	0.087
Capsule	0.983	0.996	0.982	0.982	0.982	0.711	0.917	0.476	0.921	0.321
Carpet	0.980	0.994	0.926	0.871	0.989	0.635	0.835	0.635	0.797	0.528
Grid	0.960	0.986	0.916	0.980	0.860	0.749	0.901	0.602	0.962	0.439
Hazelnut	0.999	0.999	0.971	1.000	0.943	0.863	0.902	0.500	0.923	0.343
Leather	0.997	0.998	0.898	0.814	1.000	0.696	0.892	0.786	0.740	0.837
Metal Nut	0.993	0.998	0.963	0.939	0.989	0.500	0.825	0.243	0.929	0.140
Pill	0.914	0.982	0.891	0.983	0.816	0.660	0.923	0.395	0.972	0.248
Screw	0.957	0.983	0.796	0.976	0.672	0.777	0.887	0.421	0.970	0.269
Tile	0.981	0.993	0.951	0.987	0.917	0.501	0.727	0.132	0.857	0.071
Toothbrush	0.900	0.949	0.923	0.857	1.000	0.783	0.922	0.735	0.947	0.600
Transistor	0.985	0.976	0.916	0.884	0.950	0.848	0.758	0.657	0.767	0.575
Wood	0.987	0.995	0.952	0.922	0.983	0.859	0.955	0.182	1.000	0.100
Zipper	0.954	0.985	0.962	0.966	0.958	0.787	0.940	0.513	1.000	0.345

Table 3: Image-level calibrated thresholds and confusion counts on the official test set.

Category	PatchCore (ResNet-18)					Keras CAE (Baseline)				
	Threshold	TP	FP	FN	TN	Threshold	TP	FP	FN	TN
Bottle	7.371	63	5	0	15	0.407	62	0	1	20
Cable	15.064	84	4	8	54	0.811	8	4	84	54
Capsule	6.646	107	2	2	21	0.412	35	3	74	20
Carpet	7.338	88	13	1	15	0.467	47	12	42	16
Grid	9.984	49	1	8	20	0.445	25	1	32	20
Hazelnut	14.282	66	0	4	40	0.492	24	2	46	38
Leather	6.924	92	21	0	11	0.495	77	27	15	5
Metal Nut	11.597	92	6	1	16	0.598	13	1	80	21
Pill	10.507	115	2	26	24	0.414	35	1	106	25
Screw	10.511	80	2	39	39	0.496	32	1	87	40
Tile	10.931	77	1	7	32	0.857	6	1	78	32
Toothbrush	9.525	30	5	0	7	0.669	18	1	12	11
Transistor	11.466	38	5	2	55	0.599	23	7	17	53
Wood	10.032	59	5	1	14	0.700	6	0	54	19
Zipper	7.867	114	4	5	28	0.625	41	0	78	32

Table 4: Complete stored pixel-level metrics. Pixel AUROC is retained here for conventional reference but is not used as a primary model-selection metric.

Category	PatchCore (ResNet-18)			Keras CAE (Baseline)		
	Px AUROC	**Px F1**†	**AUPIMO**	**Px AUROC**	**Px F1**†	**AUPIMO**
Bottle	0.978	0.406	0.982	0.967	0.519	0.507
Cable	0.982	0.540	0.511	0.747	0.000	0.000
Capsule	0.988	0.176	0.563	0.947	0.162	0.052
Carpet	0.988	0.169	0.796	0.949	0.202	0.003
Grid	0.970	0.200	0.506	0.972	0.270	0.077
Hazelnut	0.989	0.517	0.863	0.890	0.067	0.033
Leather	0.989	0.070	0.972	0.976	0.433	0.075
Metal Nut	0.984	0.758	0.760	0.691	0.002	0.002
Pill	0.978	0.591	0.293	0.888	0.188	0.030
Screw	0.991	0.132	0.380	0.970	0.101	0.005
Tile	0.929	0.532	0.685	0.634	0.003	0.001
Toothbrush	0.989	0.280	0.414	0.925	0.091	0.054
Transistor	0.969	0.598	0.427	0.787	0.037	0.005
Wood	0.919	0.273	0.642	0.853	0.000	0.008
Zipper	0.980	0.302	0.251	0.933	0.049	0.025

†Pixel F1 is included as a stored operating-point diagnostic, but the two implementations are not directly comparable: PatchCore uses the 99th percentile of normal-validation pixel scores, whereas the CAE reuses its normal-validation image-score threshold on the pixel error map. AUPIMO is the protocol-consistent pixel metric used for the final comparison.

Table 5: Shared evaluation protocol and stored metric provenance.

Item	**Protocol**
Image operating threshold	Category-specific 95th percentile of normal-validation image scores; frozen before official test inference.
PatchCore pixel-F1 threshold	Category-specific 99th percentile of normal-validation pixel scores.
CAE pixel-F1 threshold	Normal-validation image-score threshold reused on pixel error scores; reported only as a diagnostic.
AUPIMO false-positive range	Shared FPR from 10^{-5} to 10^{-4} on a logarithmic scale.
AUPIMO thresholds	50,000 thresholds.
Canonical pixel-map size	256×256 for both predictions and ground-truth masks.
Image PR-AUC	Trapezoidal area under the stored image precision-recall curve.

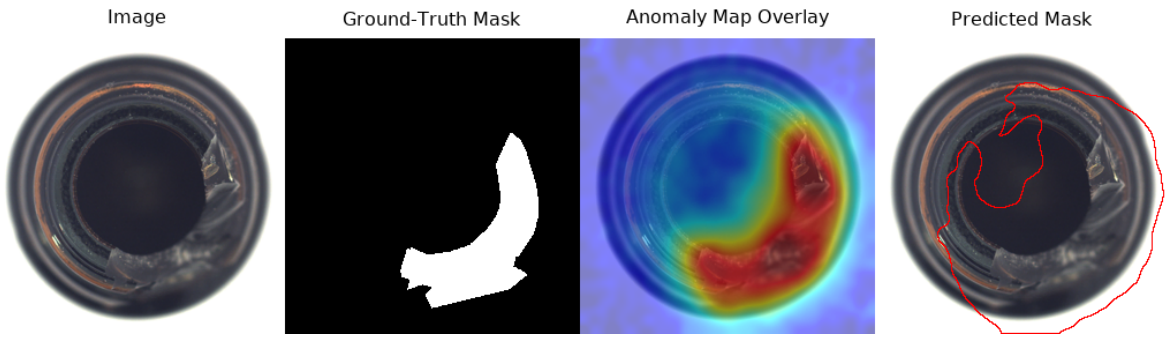
Qualitative Visualisations and Evaluation Curves

For six representative categories, we present qualitative anomaly maps for the two baseline models. We show two defective test images per category.

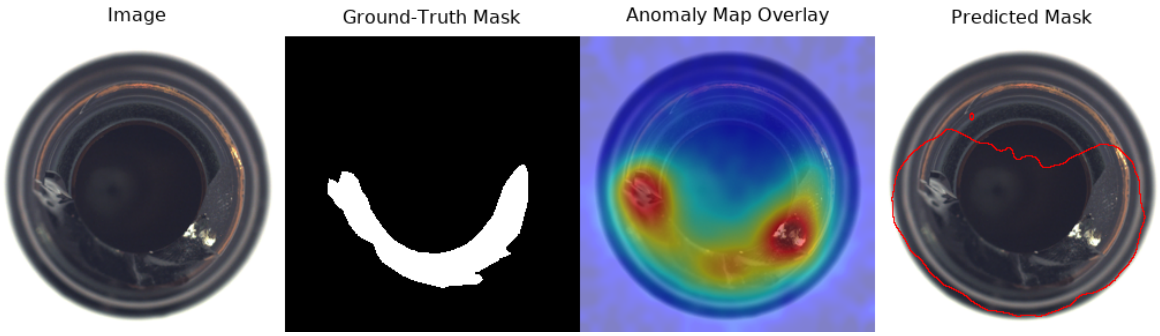
How to read the visualisations: Each baseline PatchCore example is a four-panel inference result showing the original image, the binary ground-truth mask, the continuous anomaly-map overlay, and the final predicted mask. The predicted mask uses the category-specific threshold fixed at the 99th percentile of anomaly scores from normal validation pixels; the official test masks do not influence this threshold. The baseline Keras CAE uses paired anomaly-map overlays, with the second panel adding the ground-truth defect outline. Darker colours indicate regions considered normal, while warmer, brighter colours indicate higher anomaly scores. These views distinguish the model's continuous localisation evidence from its final thresholded decision.

We also include the baseline evaluation curves for the remaining five categories; the corresponding *bottle* curves are in the main text.

Visualisations and Curves: Bottle

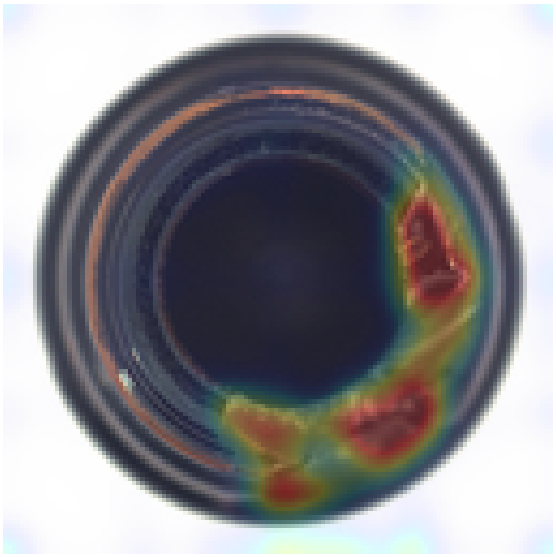


(a) Broken-large defect, sample 000.

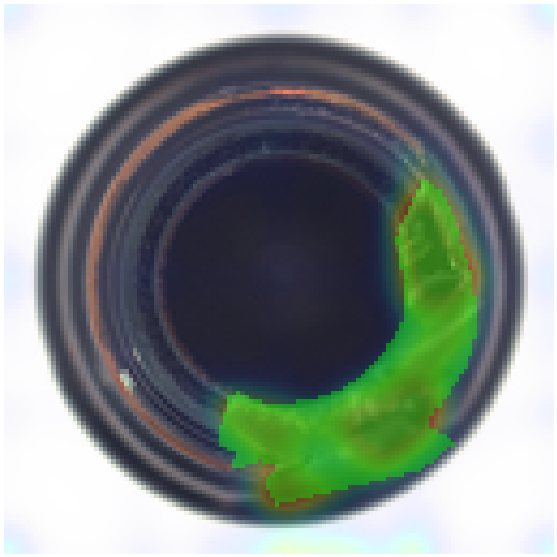


(b) Broken-large defect, sample 001.

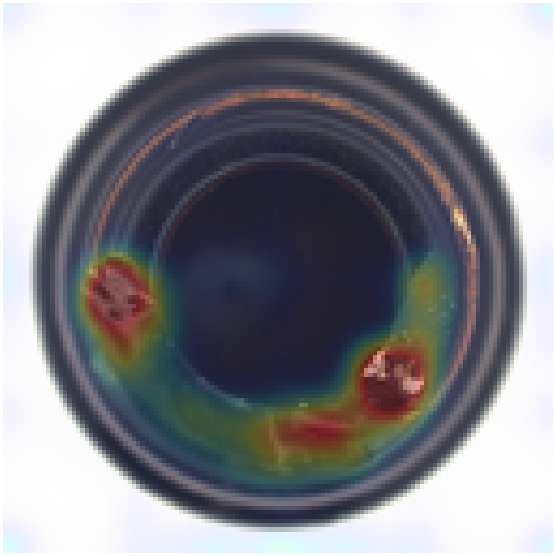
Figure 3: Four-panel qualitative results for Bottle using PatchCore (ResNet-18).



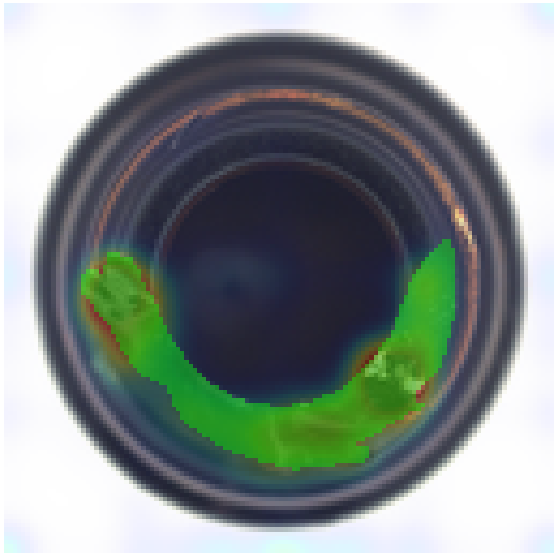
(a) Img 0: Anomaly Map Overlay



(b) Img 0: Anomaly Map + GT Outline



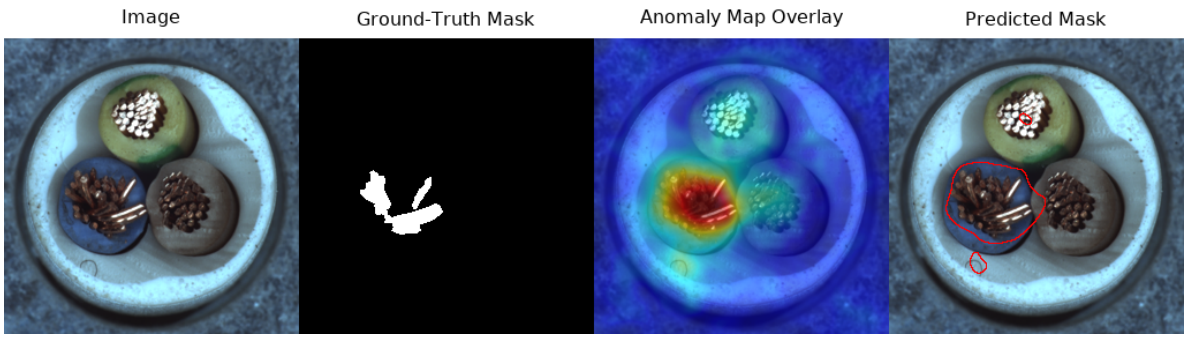
(c) Img 1: Anomaly Map Overlay



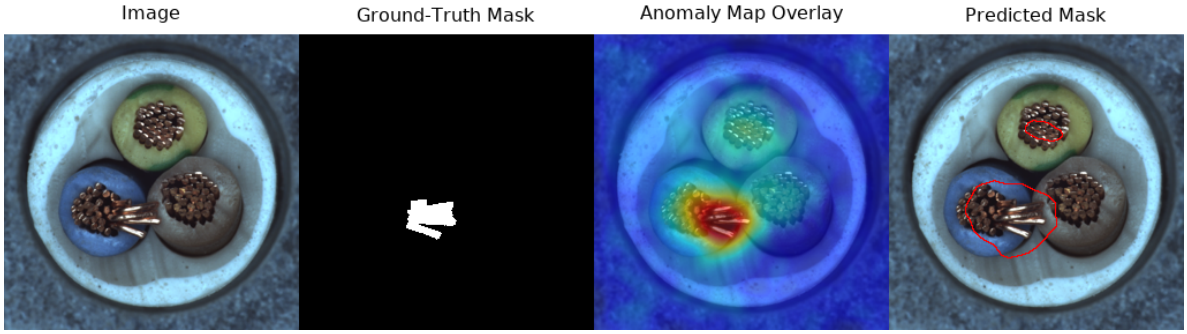
(d) Img 1: Anomaly Map + GT Outline

Figure 4: Qualitative results for Bottle using Keras CAE (Baseline).

Visualisations and Curves: Cable

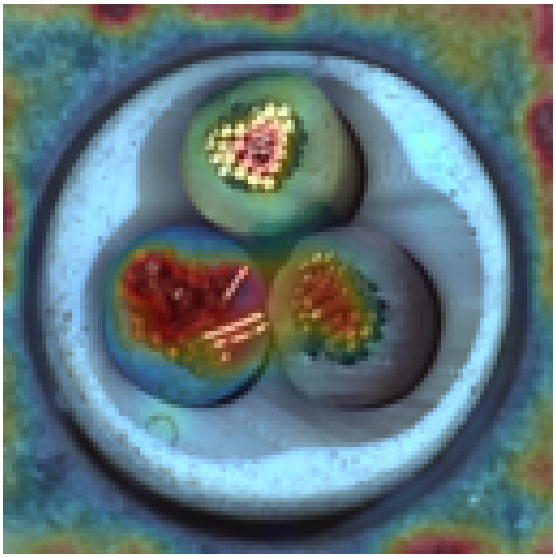


(a) Bent-wire defect, sample 000.

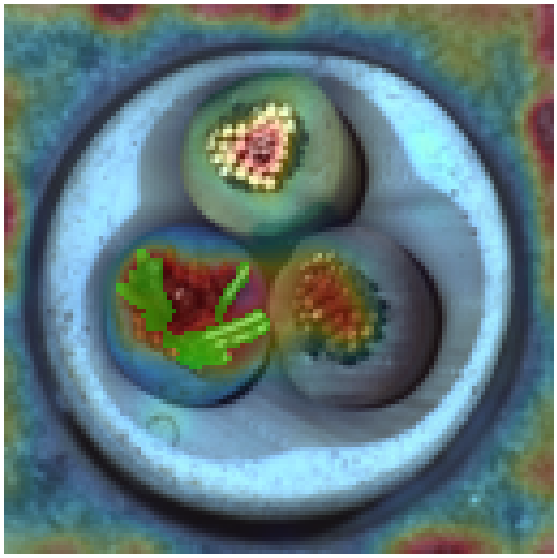


(b) Bent-wire defect, sample 001.

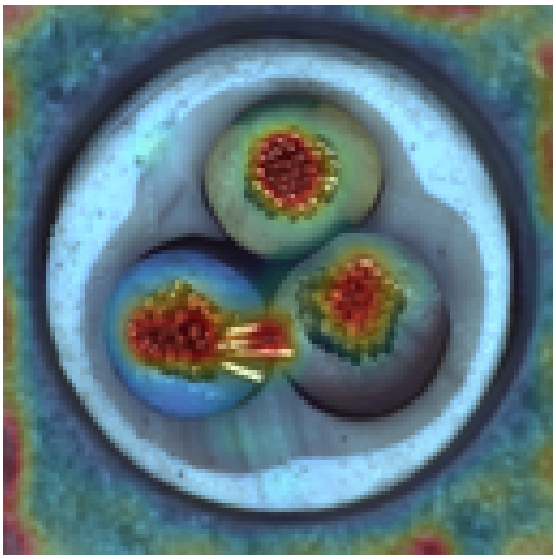
Figure 5: Four-panel qualitative results for Cable using PatchCore (ResNet-18).



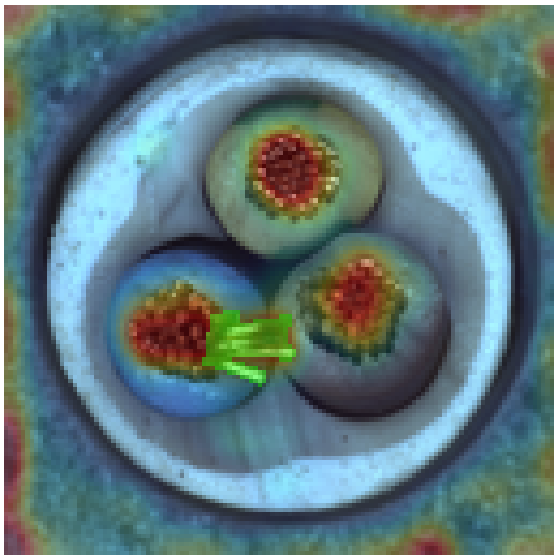
(a) Img 0: Anomaly Map Overlay



(b) Img 0: Anomaly Map + GT Outline

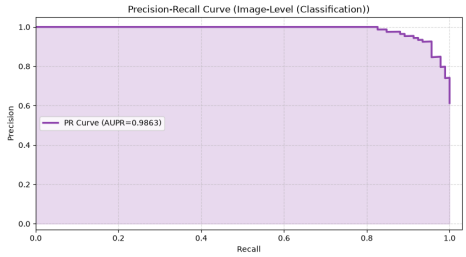


(c) Img 1: Anomaly Map Overlay

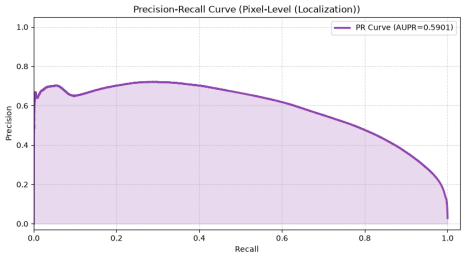


(d) Img 1: Anomaly Map + GT Outline

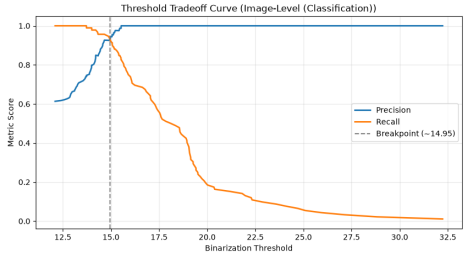
Figure 6: Qualitative results for Cable using Keras CAE (Baseline).



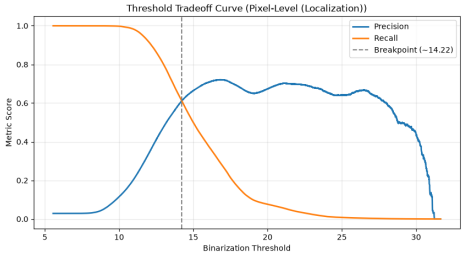
(a) Image-Level PR Curve



(b) Pixel-Level PR Curve

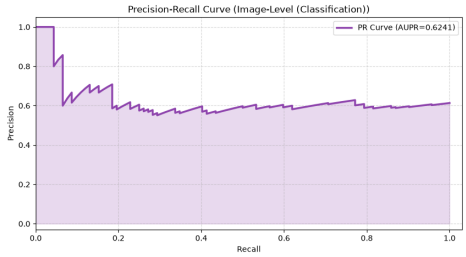


(c) Image-Level Threshold Tradeoff

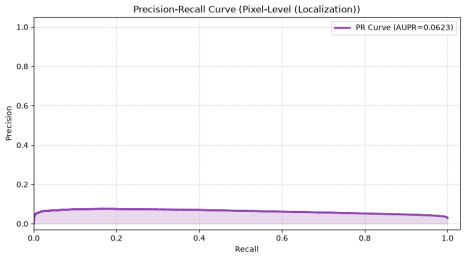


(d) Pixel-Level Threshold Tradeoff

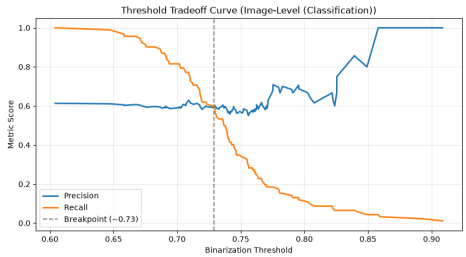
Figure 7: Evaluation curves for the *cable* category using PatchCore (ResNet-18). The pixel-level AUPIMO is 0.511.



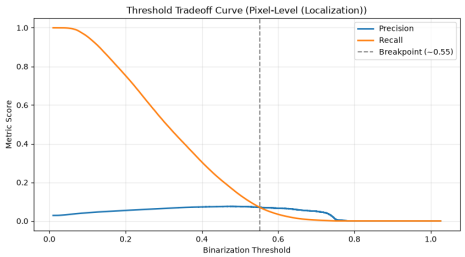
(a) Image-Level PR Curve



(b) Pixel-Level PR Curve



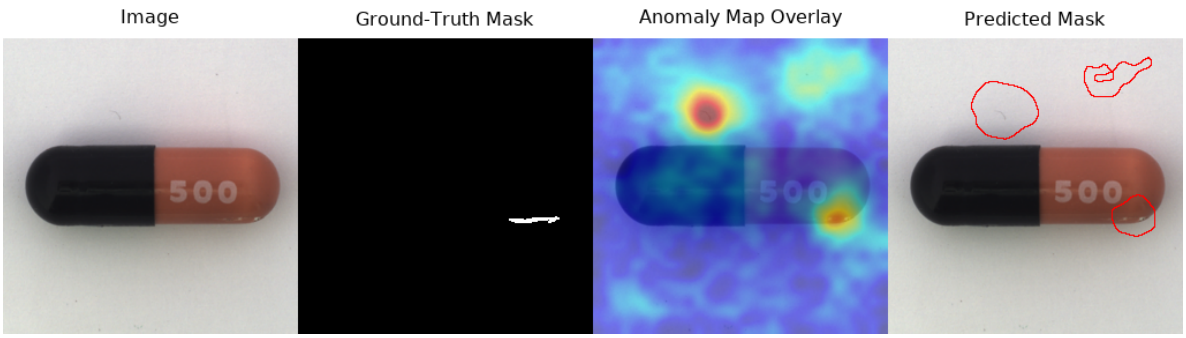
(c) Image-Level Threshold Tradeoff



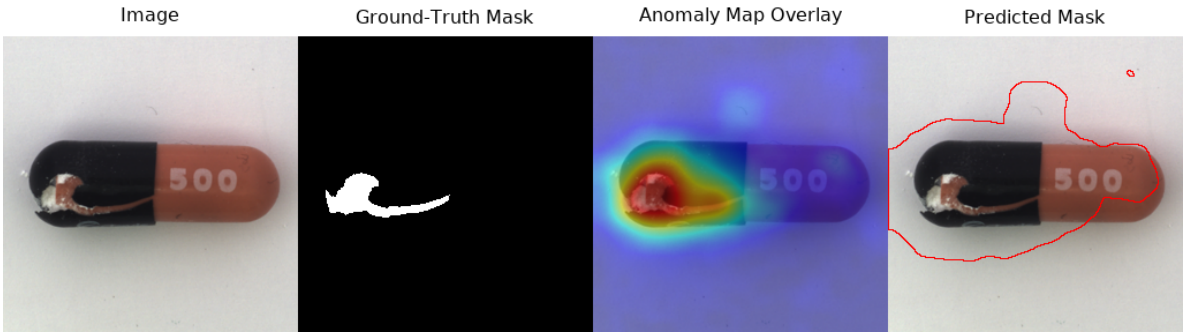
(d) Pixel-Level Threshold Tradeoff

Figure 8: Evaluation curves for the *cable* category using Keras CAE (Baseline). The pixel-level AUPIMO is 0.000.

Visualisations and Curves: Capsule



(a) Crack defect, sample 000.



(b) Crack defect, sample 001.

Figure 9: Four-panel qualitative results for Capsule using PatchCore (ResNet-18).

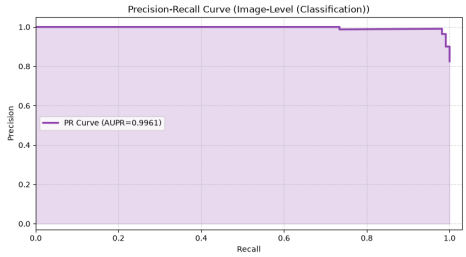
(a) Img 0: Anomaly Map Overlay

(b) Img 0: Anomaly Map + GT Outline

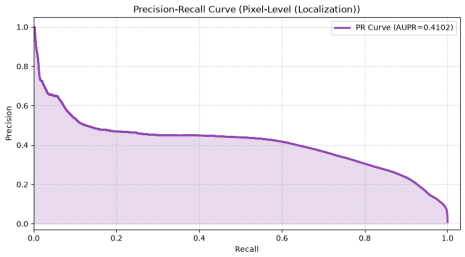
(c) Img 1: Anomaly Map Overlay

(d) Img 1: Anomaly Map + GT Outline

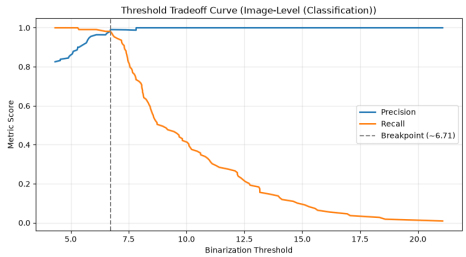
Figure 10: Qualitative results for Capsule using Keras CAE (Baseline).



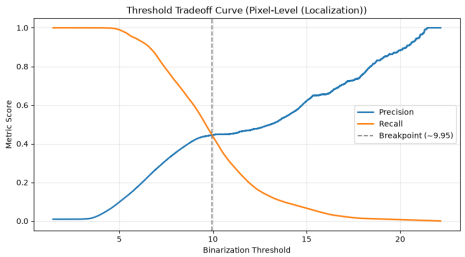
(a) Image-Level PR Curve



(b) Pixel-Level PR Curve

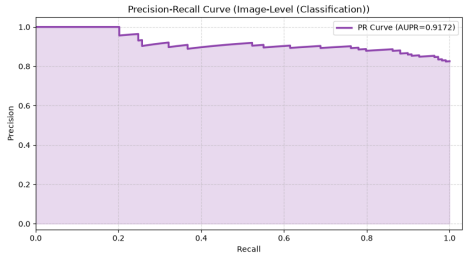


(c) Image-Level Threshold Tradeoff

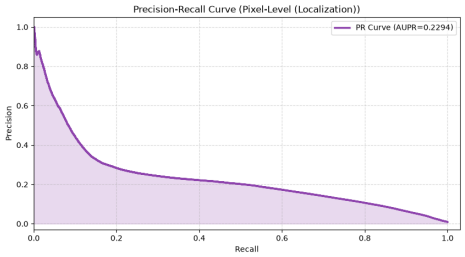


(d) Pixel-Level Threshold Tradeoff

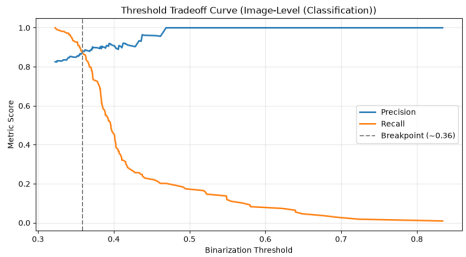
Figure 11: Evaluation curves for the *capsule* category using PatchCore (ResNet-18). The pixel-level AUPIMO is 0.563.



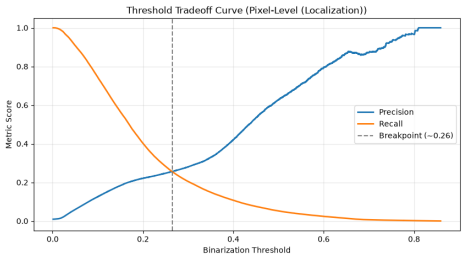
(a) Image-Level PR Curve



(b) Pixel-Level PR Curve



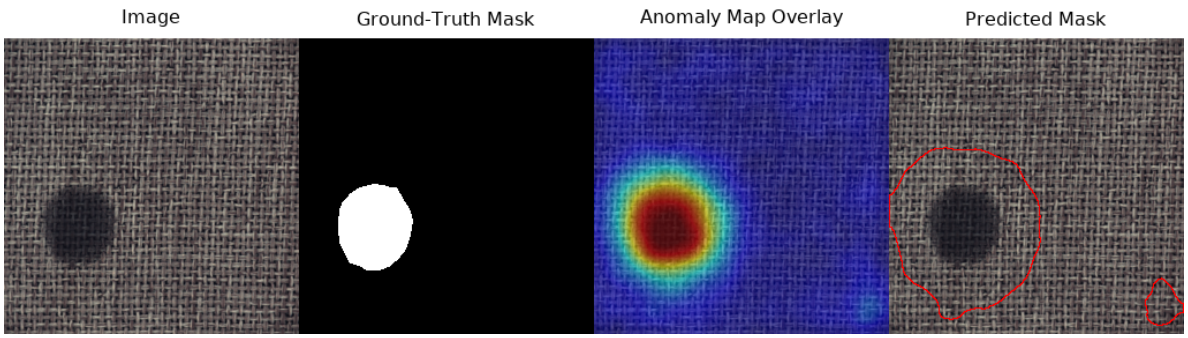
(c) Image-Level Threshold Tradeoff



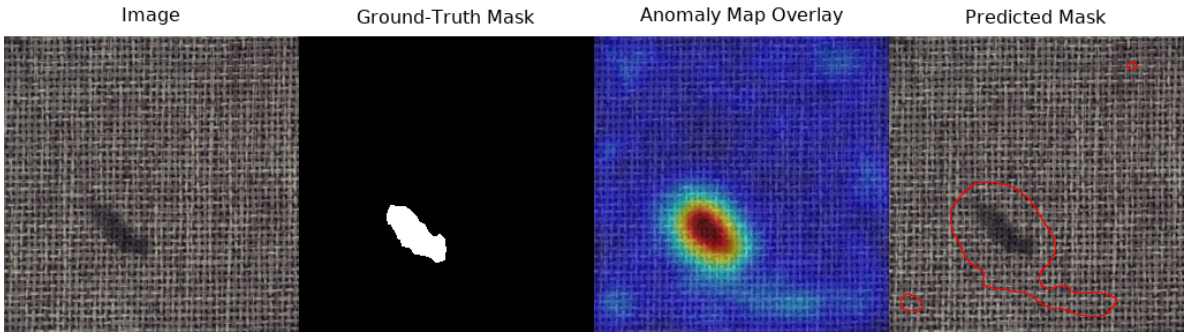
(d) Pixel-Level Threshold Tradeoff

Figure 12: Evaluation curves for the *capsule* category using Keras CAE (Baseline). The pixel-level AUPIMO is 0.052.

Visualisations and Curves: Carpet

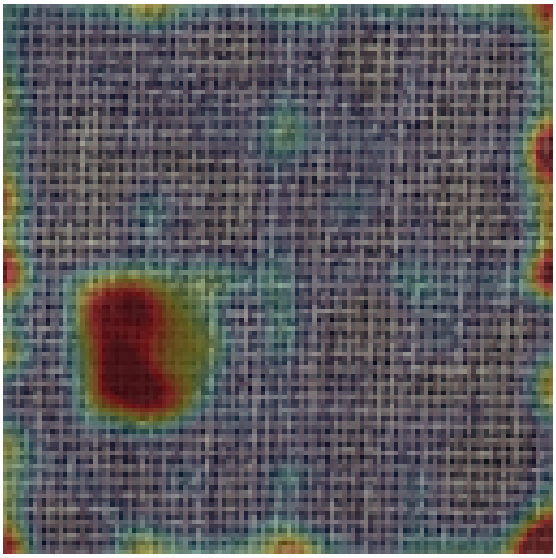


(a) Colour defect, sample 000.

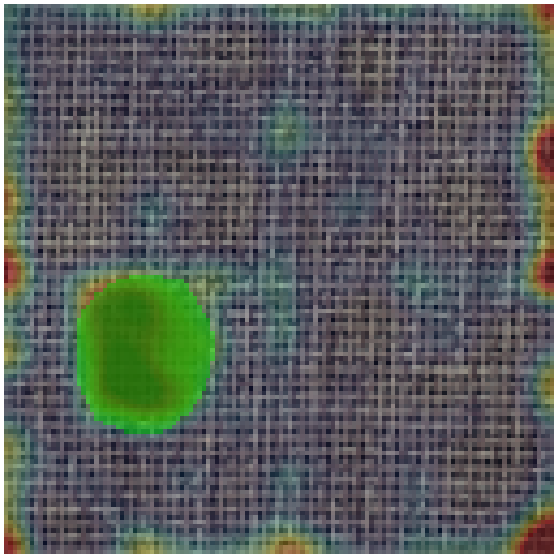


(b) Colour defect, sample 001.

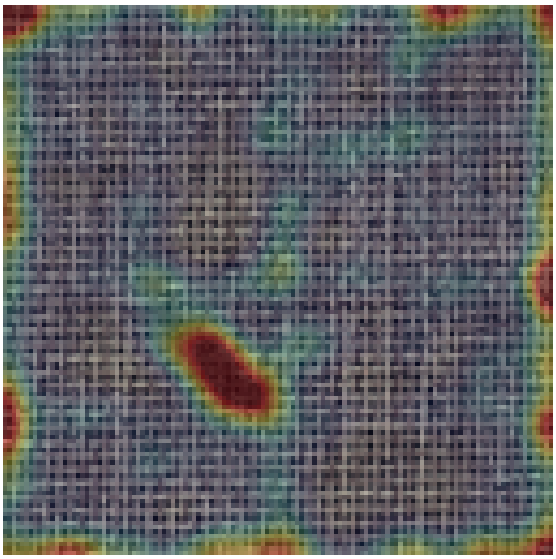
Figure 13: Four-panel qualitative results for Carpet using PatchCore (ResNet-18).



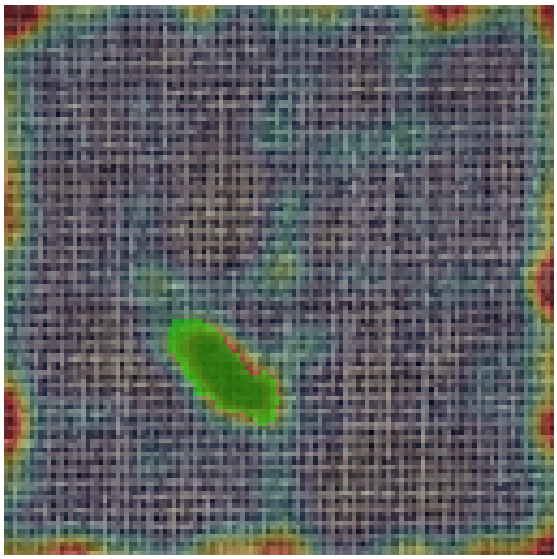
(a) Img 0: Anomaly Map Overlay



(b) Img 0: Anomaly Map + GT Outline

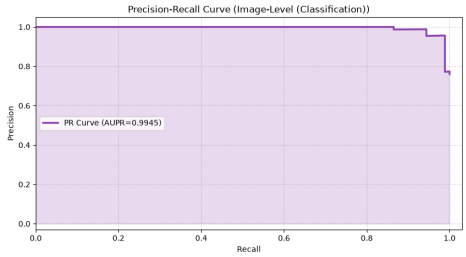


(c) Img 1: Anomaly Map Overlay

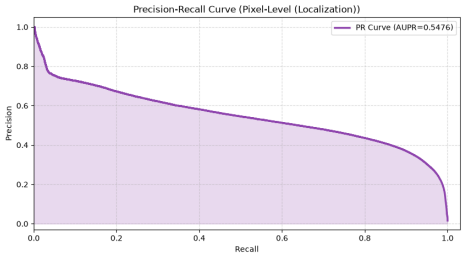


(d) Img 1: Anomaly Map + GT Outline

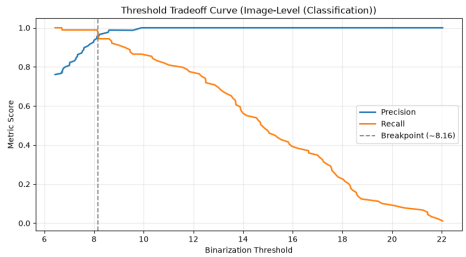
Figure 14: Qualitative results for Carpet using Keras CAE (Baseline).



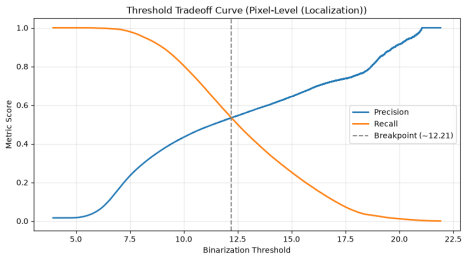
(a) Image-Level PR Curve



(b) Pixel-Level PR Curve

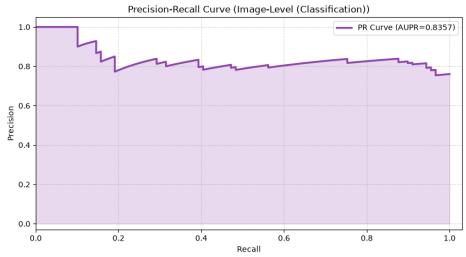


(c) Image-Level Threshold Tradeoff

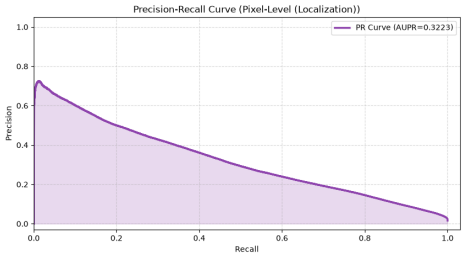


(d) Pixel-Level Threshold Tradeoff

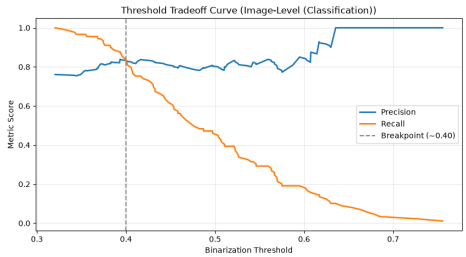
Figure 15: Evaluation curves for the *carpet* category using PatchCore (ResNet-18). The pixel-level AUPIMO is 0.796.



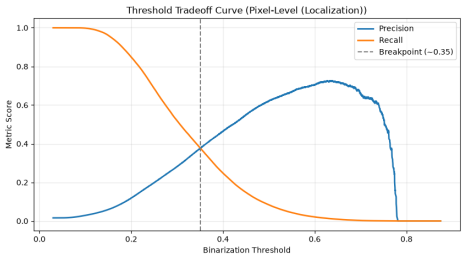
(a) Image-Level PR Curve



(b) Pixel-Level PR Curve



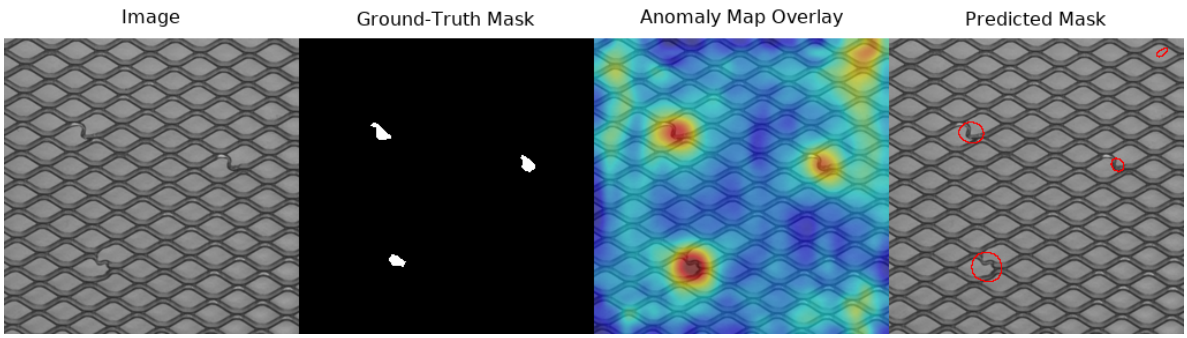
(c) Image-Level Threshold Tradeoff



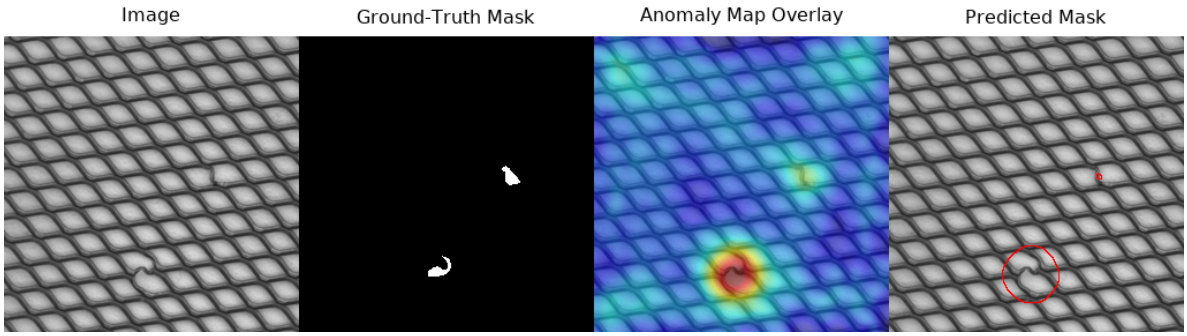
(d) Pixel-Level Threshold Tradeoff

Figure 16: Evaluation curves for the *carpet* category using Keras CAE (Baseline). The pixel-level AUPIMO is 0.003.

Visualisations and Curves: Grid

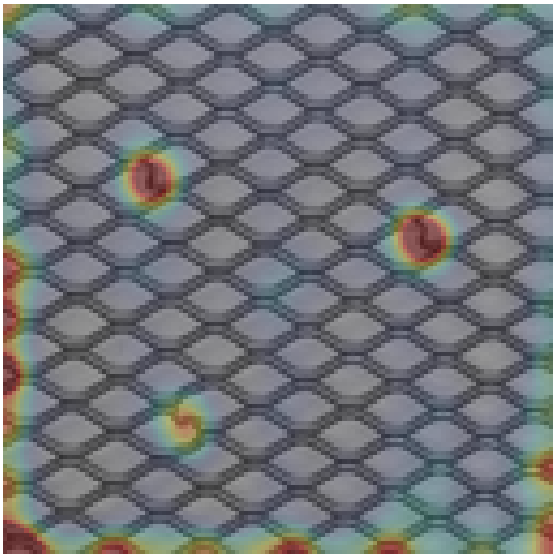


(a) Bent defect, sample 000.

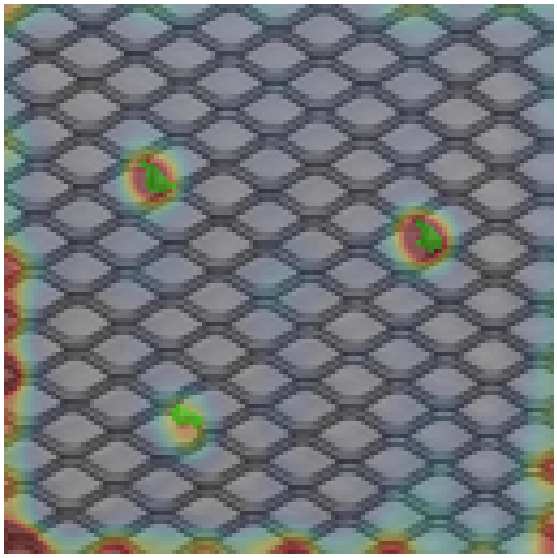


(b) Bent defect, sample 001.

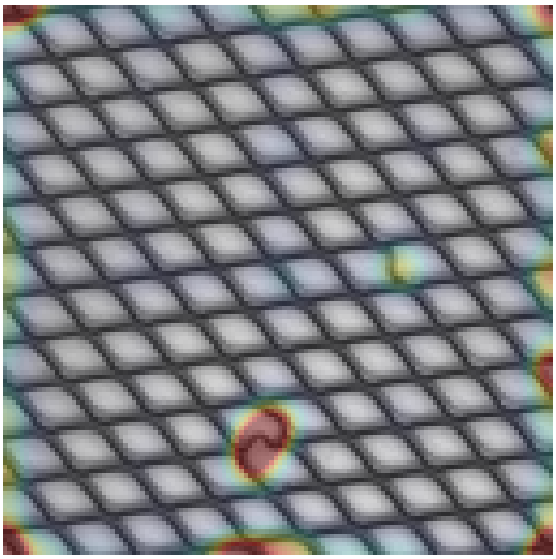
Figure 17: Four-panel qualitative results for Grid using PatchCore (ResNet-18).



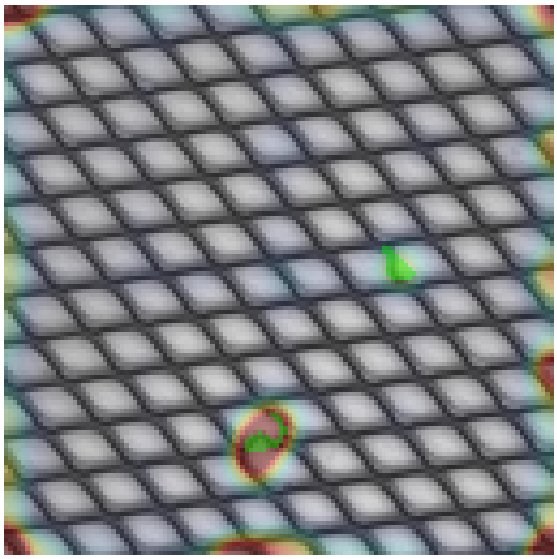
(a) Img 0: Anomaly Map Overlay



(b) Img 0: Anomaly Map + GT Outline

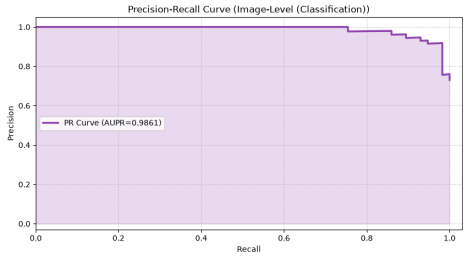


(c) Img 1: Anomaly Map Overlay

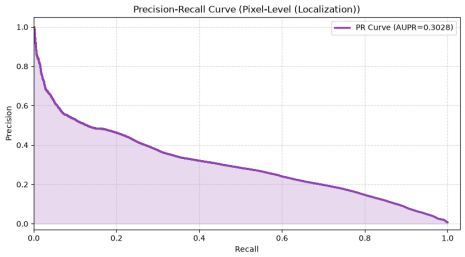


(d) Img 1: Anomaly Map + GT Outline

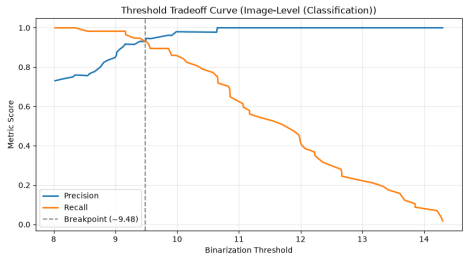
Figure 18: Qualitative results for Grid using Keras CAE (Baseline).



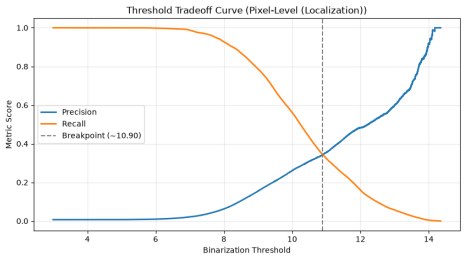
(a) Image-Level PR Curve



(b) Pixel-Level PR Curve

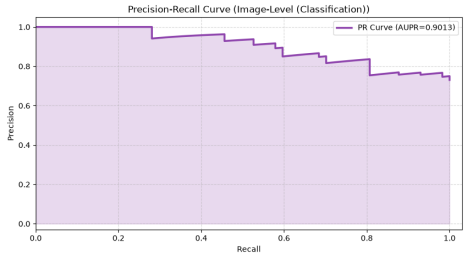


(c) Image-Level Threshold Tradeoff

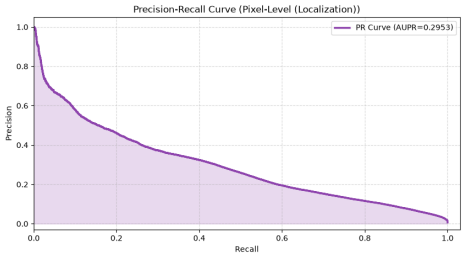


(d) Pixel-Level Threshold Tradeoff

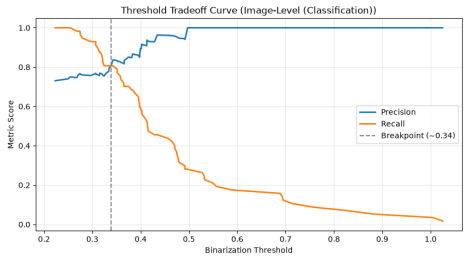
Figure 19: Evaluation curves for the *grid* category using PatchCore (ResNet-18). The pixel-level AUPIMO is 0.506.



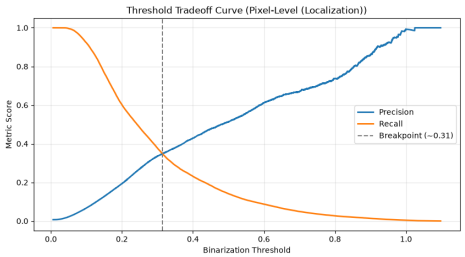
(a) Image-Level PR Curve



(b) Pixel-Level PR Curve



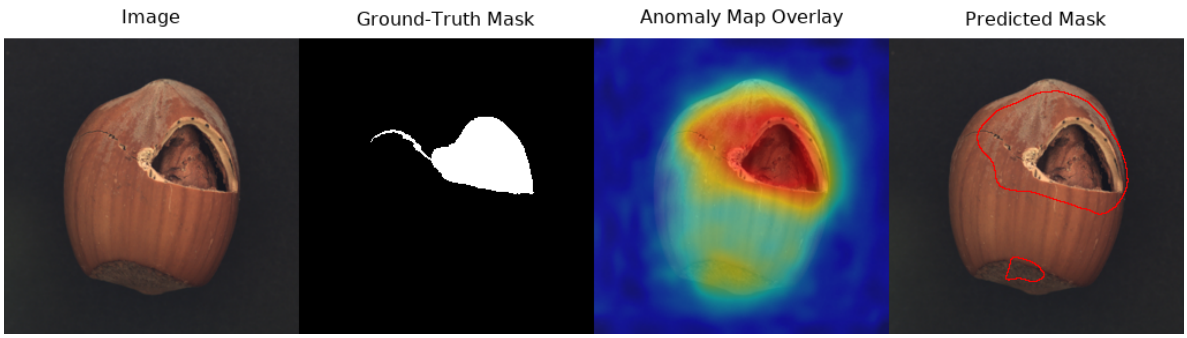
(c) Image-Level Threshold Tradeoff



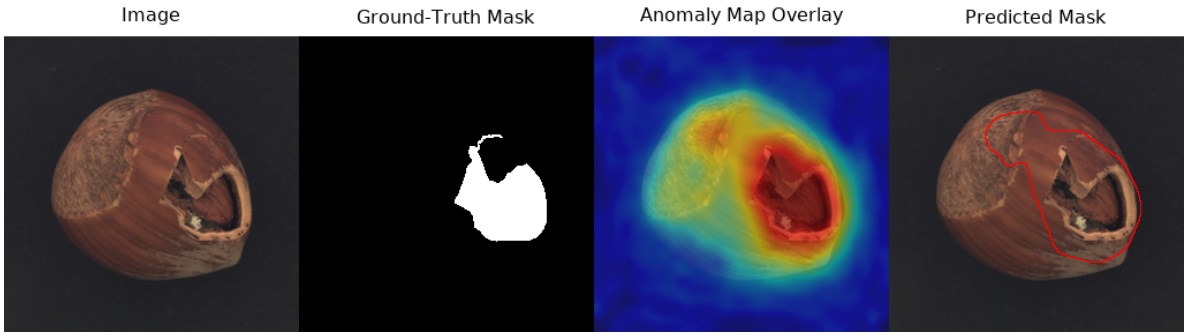
(d) Pixel-Level Threshold Tradeoff

Figure 20: Evaluation curves for the *grid* category using Keras CAE (Baseline). The pixel-level AUPIMO is 0.077.

Visualisations and Curves: Hazelnut

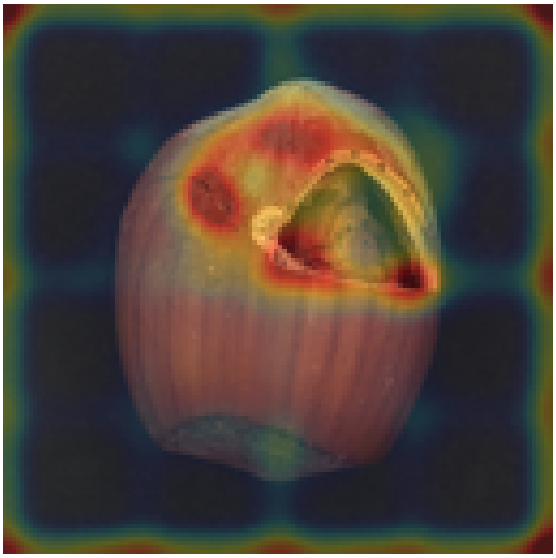


(a) Crack defect, sample 000.

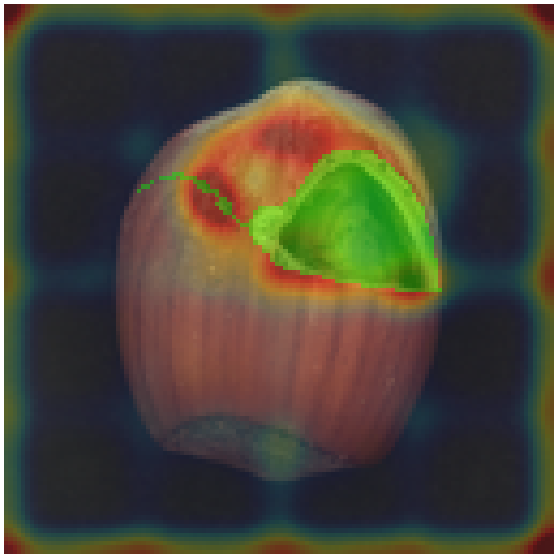


(b) Crack defect, sample 001.

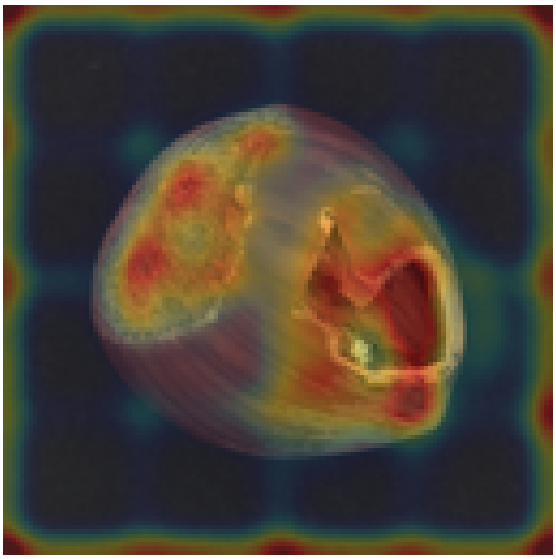
Figure 21: Four-panel qualitative results for Hazelnut using PatchCore (ResNet-18).



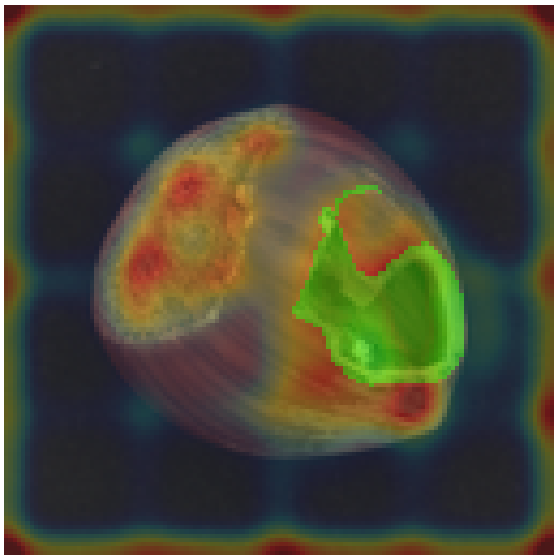
(a) Img 0: Anomaly Map Overlay



(b) Img 0: Anomaly Map + GT Outline

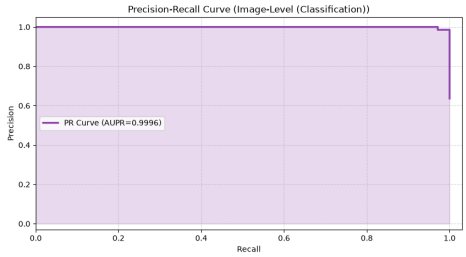


(c) Img 1: Anomaly Map Overlay

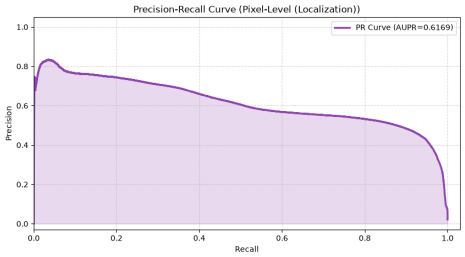


(d) Img 1: Anomaly Map + GT Outline

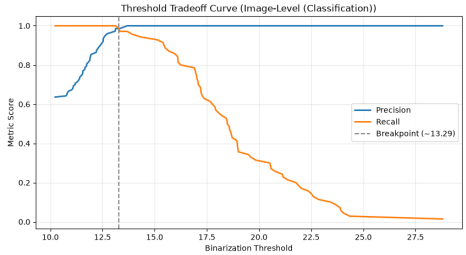
Figure 22: Qualitative results for Hazelnut using Keras CAE (Baseline).



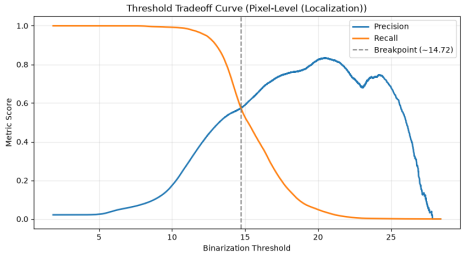
(a) Image-Level PR Curve



(b) Pixel-Level PR Curve

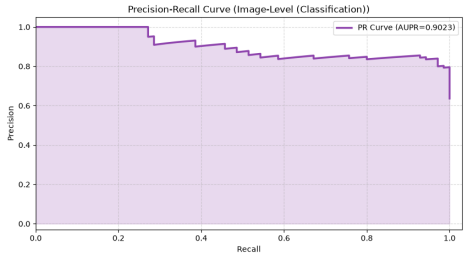


(c) Image-Level Threshold Tradeoff

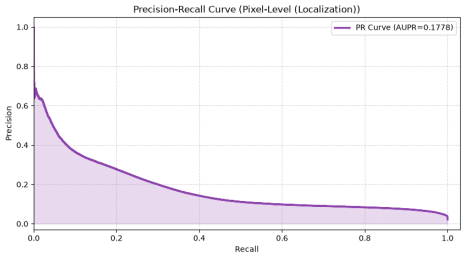


(d) Pixel-Level Threshold Tradeoff

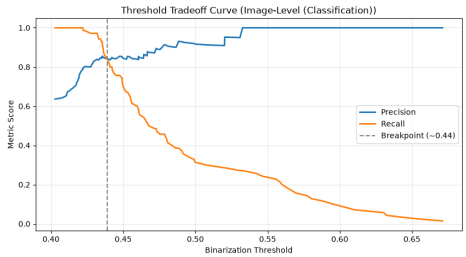
Figure 23: Evaluation curves for the *hazelnut* category using PatchCore (ResNet-18). The pixel-level AUPIMO is 0.863.



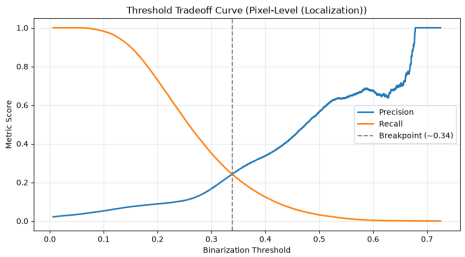
(a) Image-Level PR Curve



(b) Pixel-Level PR Curve



(c) Image-Level Threshold Tradeoff



(d) Pixel-Level Threshold Tradeoff

Figure 24: Evaluation curves for the *hazelnut* category using Keras CAE (Baseline). The pixel-level AUPIMO is 0.033.